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Review

Materials Design for Rechargeable Metal-Air Batteries

Hao-Fan Wang[1] and Qiang Xu[1,*]

In consideration of the growing problems regarding environment and energy, the development of clean and renewable electrochemical energy-storage devices has attracted much attention. Metal-air batteries with ultra-high energy density, including but not limited to Zn-air and Li-air batteries, have shown great potential for future large-scale applications. In this Review, fundamentals on the electrode reactions of metal-air batteries are firstly introduced. Current issues on achieving applicable discharge properties and rechargeability are then presented. In the following sections, the material design strategies of the air electrode, metal electrode, electrolyte, and separator to overcome these issues are summarized. The most widely investigated Zn-air and Li-air batteries are overviewed in detail, while other types of metal-air batteries including Al-air, Mg-air, and Na-air batteries are briefly discussed. Finally, summary and perspectives on the future development of metal-air batteries toward practical applications are provided.

Introduction

As increasingly severe problems of environmental pollution and resource exhaustion are brought about by the current fossil fuel-based energy system, the need for clean and renewable energy sources is becoming increasingly urgent.[1] Electricity is a green and highly efficient end-use energy. Nowadays, electrochemical energy-storage devices with various time scales and size scales have penetrated into every aspect of our daily life, including smart grids, electric vehicles, and smart mobile devices.[2] During the research history of electrochemical energy storage, the research advances of materials science have largely promoted the development and commercialization of electrochemical energy-storage devices such as Li-ion batteries.[3] With intense efforts applied to materials research, the future application of more kinds of electrochemical energy-storage devices can provide further benefit.

Metal-air batteries are a family of electrochemical cells powered by metal oxidation and oxygen reduction, exhibiting a great advantage regarding theoretical energy density, which is about 3–30 times higher than commercial Li-ion batteries.[4] Li-air batteries and Zn-air batteries are two types of metal-air batteries that have attracted most attention.[5] Li-air batteries (with Li_2O_2 as the discharge product) can deliver a superior theoretical performance of a large energy density of 11,429 W h kg^{-1} (based on the mass of Li metal), a high specific capacity of 3,860 mA h g^{-1} (based on the mass of Li metal), and a cell voltage as high as 2.96 V.[6] For Zn-air batteries the theoretical energy density is 1,350 W h kg^{-1} (based on the mass of Zn metal), still about 5-fold higher than Li-ion batteries.[7] Moreover, Zn-air batteries also have the advantage of low cost (~\$10 kW^{-1} h^{-1}) compared with Li-ion batteries (\$400–800 kW^{-1} h^{-1}).[7] Besides Li-air and Zn-air batteries, other types of metal-air batteries also have their own superiorities. For example, Al-air batteries exhibit

Progress and Potential

Metal-air batteries, powered by metal oxidation and oxygen reduction, have been intensely focused upon as promising next-generation high-energy batteries. However, the practical application of rechargeable metal-air batteries faces challenges, such as unsatisfying power density, round-trip efficiency, and Coulombic efficiency. Rational material design of the air electrode, metal electrode, electrolyte, and separator is highly desired to address these issues. This Review presents recent advances on the material design toward high-performance rechargeable metal-air batteries. Targets and principles for the material design of each part of metal-air batteries are highlighted. For future investigations, greater emphasis should be placed on achieving stable and reversible charge-discharge cycles to accelerate the development of truly rechargeable metal-air batteries.

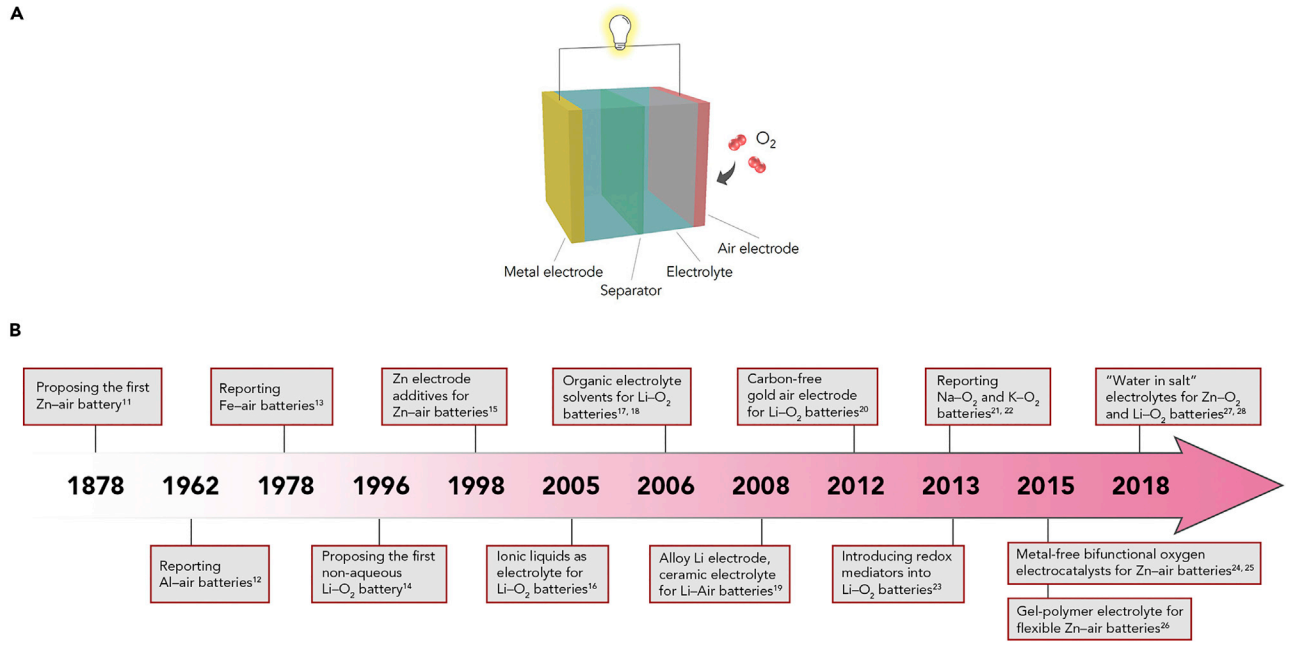


Figure 1. Introduction of Metal-Air Batteries

(A) Schematic illustration of a metal-air battery.

(B) A brief timeline of research progress in metal-air batteries.[11–28]

the highest volumetric capacity (8,040 A h L^{-1}),[8] and Na-air batteries show smaller charge overpotentials than Li-air batteries.[9] Therefore, the metal-air battery family has great potential to serve as next-generation electrochemical energy-storage devices.

Metal-air batteries usually contain four major parts: an air electrode, metal electrode, electrolyte, and separator, as illustrated in Figure 1A. For most metal-air batteries, the air electrode reactant O_2 is obtained from the ambient air rather than encapsulated in the cell. Therefore, the air electrode usually consists of only an electrocatalyst layer to reduce the electrode overpotential and a gas-diffusion layer (GDL) to enhance the oxygen diffusion between the ambient air and the catalyst surface. The commonly investigated metals for the metal electrode include zinc, lithium, aluminum, magnesium, and sodium.[10] For highly active metals that are unstable in aqueous systems, such as Li, Na, and K, a non-aqueous aprotic electrolyte is usually required, while for relatively inactive metals such as Zn, Al, Mg, and Fe, alkaline aqueous electrolytes are widely employed.[4] The separator is an optional component in a metal-air battery used to separate two different electrolytes, obstruct some mass transportation processes between the electrodes, and prevent the short-circuit caused by metal dendrites. Since the invention of the first primary Zn-air battery in 1878,[11] much progress has been made in the research on metal-air batteries (Figure 1B).[11–28] However, these batteries still face problems that impede their large-scale industrial application. To realize the practical application of metal-air batteries, the material design and regulation of each component should be considered.

In this review we first introduce the mechanism, evaluation parameters, and existing problems of metal-air batteries. Based on the fundamental understandings of metal-air batteries, representative materials design strategies to overcome the challenges

[1]AIST-Kyoto University Chemical Energy Materials Open Innovation Laboratory (ChEM-OIL), National Institute of Advanced Industrial Science and Technology (AIST), Yoshida, Sakyo-ku, Kyoto 606-8501, Japan

*Correspondence: q.xu@aist.go.jp

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and improve the metal-air battery performance are described in the order air electrode, metal electrode, electrolyte, and separator materials. Finally, a summary of materials design for metal-air batteries and a perspective for the future challenges of these batteries are presented.

Fundamentals and Challenges of Metal-Air Batteries

Electrode Reactions

The electrode reactions of rechargeable metal-air batteries vary with the metal electrodes and the type of electrolytes. Generally, for metal-air batteries with aqueous electrolytes the electrode reactions are in the following form:

$$\text{Metal electrode: } M \leftrightarrow M^{n+} + ne^-, \quad \text{(Equation 1)}$$

$$\text{Air electrode: } O_2 + 4e^- \leftrightarrow 2H_2O + 4OH^-, \quad \text{(Equation 2)}$$

where M stands for the metal (Zn, Al, Mg, Fe, etc.) and n represents the metal ion charge number. The metal ions generated on the metal electrode during the discharge process may further react with OH^- in the alkaline electrolyte. On the air electrode, the conversions between oxygen and water take place during the discharge and charge processes, which have been widely investigated as oxygen reduction reaction (ORR, discharge) and oxygen evolution reaction (OER, charge) for the applications in fuel cells, water splitting, and so forth. For Li-, Na-, and K-air batteries with aprotic electrolytes, oxygen reacts with metal ions on the air electrode, and the discharge product on the air electrode can be metal superoxides or peroxides. The electrode reactions are as follows:

$$\text{Metal electrode: } M \leftrightarrow M^+ + e^-, \quad \text{(Equation 3)}$$

$$\text{Air electrode: } xM^+ + O_2 + xe^- \leftrightarrow M_xO_2 \ (x = 1 \text{ or } 2). \quad \text{(Equation 4)}$$

The oxygen molecules are first reduced to superoxide ion and then bound with metal ions. However, according to the hard-soft acid-base theory (HSAB), Li^+ with a small ionic radius is a hard Lewis acid, which is unstable with O_2^-.[29] Therefore, the Li superoxides will disproportionate to form Li_2O_2, the main discharge product for Li-air batteries, while for Na^+ and K^+ with increasing ionic radius, the combination with O_2^- becomes more stable and the proportion of superoxides in the discharge product increases.[4] The peroxides and superoxides generated during the discharge process deposit on the air electrode, and decompose to metal ion and oxygen when charging. The air electrode ORR and OER processes significantly distinguish from those in aqueous electrolytes. Moreover, as the Li, Na, and K metals are sensitive to the H_2O and CO_2 in air, metal-air batteries with these anodes are usually operated in pure O_2 and are denoted as Li-O_2, Na-O_2, and K-O_2 batteries.

Evaluation Parameters and Current Issues

The performance evaluation of rechargeable metal-air batteries can be divided into two aspects, the discharge properties and the rechargeability. So far, most of the metal-air batteries under study still have only very limited practical application areas. How to improve the metal-air battery performance regarding both discharge properties and rechargeability is still a huge challenge. Therefore, it is crucial for promoting the further development of the metal-air batteries to study the problems and challenges in these batteries from the perspective of materials science, and look for solutions through the material design of air electrode, metal electrode, electrolyte, and separator materials. The main challenges for improving the discharge properties and the rechargeability of metal-air batteries are discussed here.

The discharge properties include the polarization performance and the capacity. The polarization performance gives information about how much the current density can be reached at a specific discharge voltage, and also reflects the peak power density that a metal-air battery can provide. The polarization performance of metal-air batteries depends on the overpotential and the reaction rate of air electrode and metal electrode reactions. The reduction of oxygen on the air electrode is sluggish in kinetics, and is considered to be the rate-determining step. Therefore, efficient electrocatalysts are required on the air electrode to reduce the oxygen reduction overpotential, thus achieving a high reaction rate.[30,31] Besides the reaction kinetics, the mass transport is also a critical factor in affecting the reaction rate.[7] Rational material designs are needed to reduce the mass transport resistance, including employing a GDL on the air electrode, regulating the pore structure of electrodes, and preventing the deposition of insulating products/by-products on the electrodes.

As the air electrode reactant O_2 is achieved from ambient air and is nearly infinite, the theoretical capacity of a metal-air battery is determined by the amount of metal encapsulated in the battery as the metal electrode. However, there are also some factors that limit the practical capacity of a metal-air battery. For aqueous metal-air batteries, passivation of the metal electrode could occur as the metal electrode is being coated by an insulating metal oxide film during the discharge process, resulting in termination of the discharge process.[10] For non-aqueous metal-air batteries, there is another cause for the degradation of capacity: the limited storage of metal peroxides or superoxides in the air electrode.[5] To minimize the capacity loss of metal-air batteries, material design strategies such as the modification of metal electrodes, the introduction of additives in the electrolytes, and the pore-structure design of air electrode electrocatalysts are under development.

Typical parameters to describe the rechargeability of metal-air batteries include round-trip efficiency, Coulombic efficiency, and cycle life. The round-trip efficiency is determined by the ratio of the energy released during discharge and the energy needed in the recharge process, representing the energy-utilizing efficiency of a metal-air battery. It is usually characterized by dividing the discharge voltage by the charge voltage obtained at a constant current density. The round-trip efficiency depends on the overpotential of both discharge and charge reactions. It requires the air electrode electrocatalyst to be bifunctional toward the conversions between oxygen and water/peroxides/superoxides. The mass transport during the discharge and charge process should also be considered. Note that in the charge process of non-aqueous metal-air batteries, the decomposition of peroxides and superoxides on the air electrode needs a solid-solid contact of the reactant and the electrocatalyst. To lower the overpotential induced by the insufficient solid-solid contact, material design methods such as regulating the air electrode structure and adding redox mediators in the electrolyte have been proposed.[30,32]

The Coulombic efficiency is defined as the ratio of charge capacity and discharge capacity in a full discharge-recharge cycle, representing the capacity loss after a cycle. The Coulombic efficiency of a battery must be higher than 99.98% in order to retain 80% of the initial capacity after 1,000 cycles.[33] However, metal-air batteries always suffer from irreversible consumption of metal electrodes and parasitic reactions, resulting in low Coulombic efficiency.[6] The low Coulombic efficiency of metal-air batteries that have been studied so far strongly limits their applications as truly rechargeable batteries in mobile electronics and electric vehicles. Optimizations

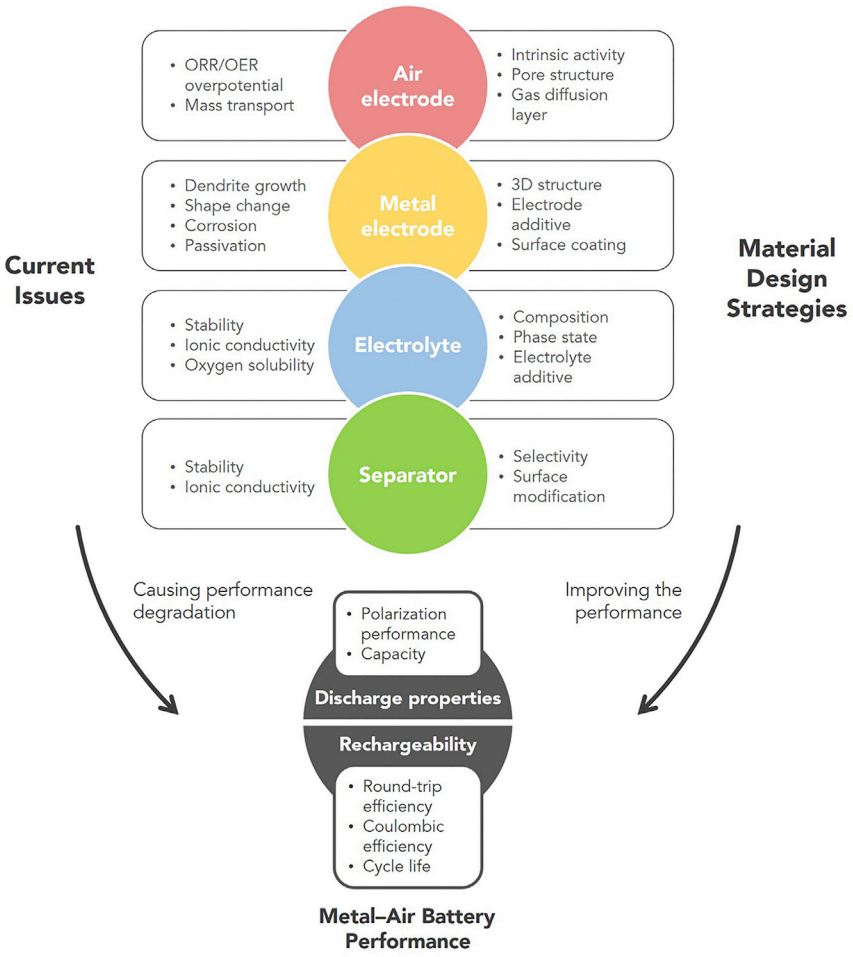


Figure 2. Summarization of Evaluation Parameters, Material Challenges, and Material Design Strategies for Metal-Air Batteries

of the electrode, electrolyte, and separator materials have been studied to mitigate this problem.[6,10]

The cycle life of a metal-air battery relies on high Coulombic efficiency, and the stability of all the parameters discussed above. A long cycle life requires the stable structure and composition of all the components in the battery. Some common issues in metal-air batteries, such as the dendrites and shape change of metal electrodes, the deactivation of electrocatalysts, and the decomposition of electrolytes, could cause shortening of the cycle life.[6,34] Corresponding modifications of materials should also be addressed.

In summary, the problems of metal-air battery performance mainly originate from the materials side. Through rational material designs these problems can be mitigated and the battery performance improved, as summarized in Figure 2. With extensive investigation, methodologies to address the above issues have been studied, especially on the design of air electrode, metal electrode, electrolyte, and separator materials, which will be summarized in this review. Considering that previous research on metal-air batteries mainly focused on Zn-air and Li-air batteries,[5] these two systems will be chosen as representatives for aqueous and non-aqueous metal-air batteries in the following discussion.

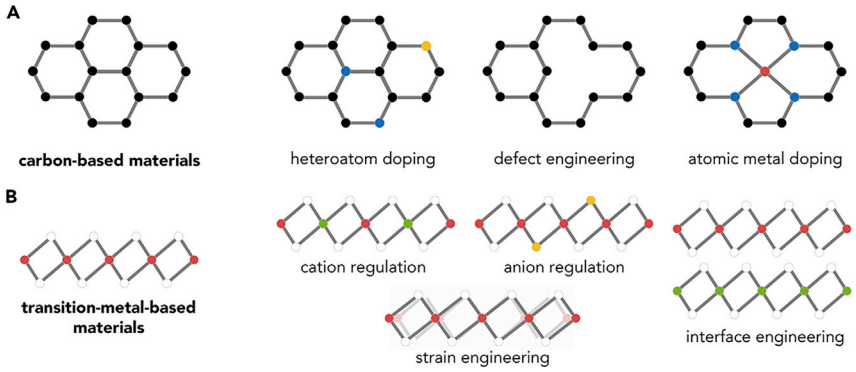


Figure 3. Material Design Strategies for Improving the Intrinsic Activity of ORR/OER Electrocatalysts in Aqueous Metal-Air Batteries

(A) Material design strategies for carbon-based materials.

(B) Material design strategies for transition-metal-based materials.

Material Design of Air Electrodes

Electrocatalyst Layer

Oxygen electrocatalysts on the air electrode of metal-air batteries play a key role in achieving high power density and high round-trip efficiency. In recent years, numerous research works and reviews on metal-air batteries placed their emphasis on the design and synthesis of oxygen electrocatalysts.[35,36] As mentioned earlier, aqueous and non-aqueous metal-air batteries have distinct air electrode reactions. Therefore, electrocatalysts for these two types of metal-air batteries will be introduced separately.

Electrocatalysts for Aqueous Metal-Air Batteries. Aqueous metal-air batteries need efficient ORR/OER bifunctional electrocatalysts on the air electrode to lower the overpotential of both the discharge and charge process. The main difficulty in developing an ORR/OER bifunctional electrocatalyst is that ORR and OER have different requirements for electrocatalysts. ORR and OER are multi-step reactions, involving the adsorption of various kinds of oxygen-containing species on the surface of electrocatalysts. Analyses of the mechanism of ORR and OER revealed that the desired adsorption energies of the oxygen-containing species on electrocatalysts to achieve the best ORR or OER activity are usually different.[37] Recently, much effort has been made to develop high-performance ORR/OER bifunctional electrocatalysts through rational structure and composition regulation of various catalysis materials.[7] Most reported materials for ORR and OER bifunctional electrocatalysis can be classified into two categories: carbon-based materials and transition-metal-based materials.[7,24,25,38–40] The common regulation strategies of these catalytic materials for achieving high intrinsic activity are illustrated in Figure 3.

For electrocatalysts with high intrinsic activity, fast gas, electrolyte, and electron transport are also desired for achieving high apparent activity. Through the combination of high intrinsic activity and favorable microstructures, the electrocatalysts employed in metal-air batteries could lower the discharge and charge overpotentials, leading to high power density and round-trip efficiency. Recently, an approach on the material design of metal-organic framework (MOF)-derived air electrode electrocatalysts was proposed by Zou et al.[41] to obtain high power density in Zn-air batteries. Cobalt nanoparticle-embedded N-doped carbon nanotubes (CNTs) grown on carbon fibers (NCo@CNT-NF700) were obtained through the pyrolysis of single-crystal cobalt MOF-74 nanotubes and dicyandiamide (DCDA) as additional

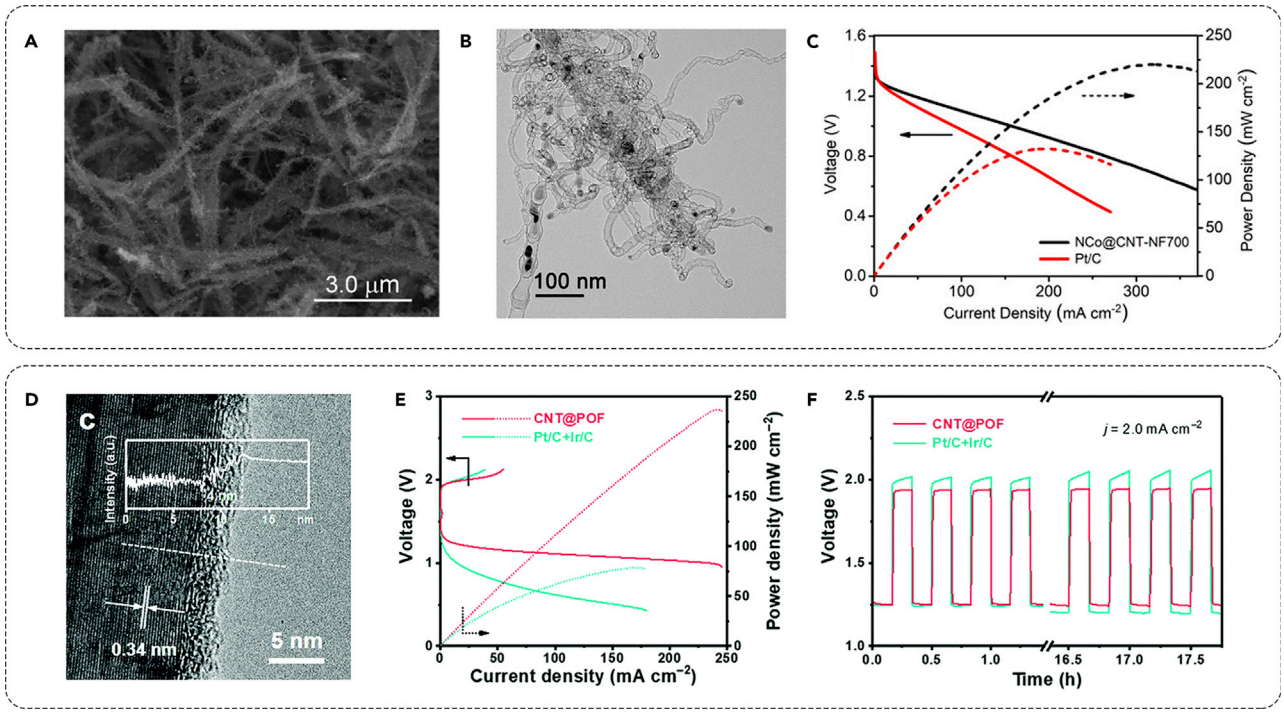


Figure 4. Material Design of Electrocatalysts in Aqueous Metal-Air Batteries

(A and B) SEM (A) and TEM (B) images of NCo@CNT-NF700.

(C) Polarization curve and corresponding power density plot of the primary Zn-air battery using NCo@CNT-NF700 and Pt/C as the air electrode catalyst. Reproduced with permission from Zou et al.[41] Copyright 2018 American Chemical Society.

(D) High-resolution TEM images of CNT@POF. Inset: the contrast profile along the white dashed line.

(E) Polarization curve and corresponding power density plot of the rechargeable Zn-air battery using CNT@POF and Pr/C+Ir/C as the air electrode catalyst.

(F) Galvanostatic discharge-charge cycling curves at 2.0 mA cm^{-2} of CNT@POF and Pt/C+Ir/C. Reproduced with permission from Li et al.[42] Copyright 2018, Royal Society of Chemistry.

carbon and nitrogen source at 700°C. A three-dimensional (3D) hierarchical structure was observed by scanning electron microscopy (SEM) and transmission electron microscopy (TEM) characterizations (Figures 4A and 4B). Co nanoparticles and N-doped carbon provide the electrocatalytic ORR activity, and the network constructed by carbon fibers and carbon nanotubes ensures the high electrical conductivity and porous structure of Co-MOF-74-NT700. As a result, Co-MOF-74-NT700 exhibited an outstanding electrocatalytic ORR performance with a half-wave potential of 0.87 V versus reversible hydrogen electrode, only 7 mV lower than a Pt/C electrocatalyst. When applied into a Zn-air battery, a high peak power density of 220 mW cm^{-2} can be achieved, shown by the polarization curve (Figure 4C). Thus, improvements on the metal-air battery discharge performance were realized by the design of efficient air electrode ORR electrocatalysts.

The development of bifunctional ORR/OER electrocatalysts can benefit both discharge and charge processes of metal-air batteries. For example, Li et al.[42] reported a hybrid material of cobalt-coordinated porphyrin covalent organic framework (POF)-coated CNTs, named CNT@POF, as a bifunctional electrocatalyst for Zn-air batteries. Thin POF layers coaxially wrap the surface of the CNTs, facilitating fast electron transfer (Figure 4D). The coating of POF also leads to a superb hydrophilic surface, which ensures the accessibility of electrolytes. Through the rational design of the CNT@POF electrocatalyst, Zn-air batteries with CNT@POF as the air

electrode electrocatalyst exhibited satisfactory discharge and charge performances. For discharging, a peak power density of 237 mW cm^{-2} was revealed by the polarization curve (Figure 4E). In respect of cycling properties, the galvanostatic discharge-charge cycling test at 2.0 mA cm^{-2} (Figure 4F) exhibited an initial discharge and charge voltage of 1.25 V and 1.96 V, respectively, with a voltage gap of 0.71 V and a round-trip efficiency as high as 63.8%. Moreover, no obvious performance degradation was observed after 200 cycles (over 66 h).

For other types of aqueous metal-air batteries such as Al-air and Mg-air batteries, the air electrode reactions are the same as those of Zn-air batteries. Therefore, these aqueous metal-air batteries share the same electrocatalyst design principles. For example, Cheng et al.[43] designed an electrocatalyst with mesoporous structure and FeN_x as active site for ORR, which exhibited superior Mg-air battery performance to the Pt/C electrocatalyst. In a word, the design of air electrode electrocatalysts for aqueous metal-air batteries mainly focuses on the fast conversions between oxygen and water with low overpotential, thus improving the power density and energy efficiency.

Electrocatalysts for Non-aqueous Metal-Air Batteries. The air electrode reactions of non-aqueous metal-air batteries are different from those of aqueous metal-air batteries, as mentioned above. Therefore, the material design of air electrode electrocatalysts for non-aqueous metal-air batteries needs different strategies to boost the air electrode reactions. Taking Li-air batteries as an example, the two-electron reactions of the formation and decomposition of Li_2O_2 contain fewer steps than four-electron aqueous ORR and OER. However, the conversion between O_2 and Li_2O_2 is still sluggish due to the complicated intermediates and multiple side reactions.[44] Moreover, the limited storage of Li_2O_2 in the air electrode and the insulating nature of Li_2O_2 could also affect the capacity and recharge properties.[45] The problems of air electrode reactions can be mitigated by the material regulations on air electrode electrocatalysts and electrolytes[6,30,46] (for the design of electrolytes, see Material Design of Electrolytes). In this section, materials for non-aqueous metal-air battery electrocatalysts are introduced from the aspect of active material selection and pore-structure design.

Carbon materials have been widely investigated as air electrode electrocatalysts and supports for Li-air batteries, for their high electrical conductivity, large specific surface area, and porous structure, which are favorable for achieving high discharge capacity and rate capability.[38,47] However, carbon materials suffer from their low activity on Li_2O_2 decomposition and poor stability at high charge potentials, resulting in inferior round-trip efficiency and cycling stability.[44,47] Therefore, much attention has been paid to non-carbon Li-air battery air electrode materials in recent years. Materials including precious metals[20,48,49] and transition metal oxides[50–53] have proved their activity on aprotic oxygen reduction and evolution reactions. Similar to aqueous metal-air batteries, structure and composition regulation strategies can be applied on the electrocatalysts to enhance the performance of non-aqueous metal-air batteries. Besides improving the activity for ORR and OER, tuning the morphology of discharge products is also an important target in the design of air electrode electrocatalysts for non-aqueous metal-air batteries.

A multifaceted design of Co_3O_4 electrocatalyst on surface, interface, and doping was recently reported by Gao et al.[54] Co_3O_4@Co_3O_4/Ag hybrids with a yolk-shell structure, as illustrated in Figure 5A, were synthesized by a synchronous reduction method. On the porous Co_3O_4 shell, part of the Ag is doped into the crystal lattice

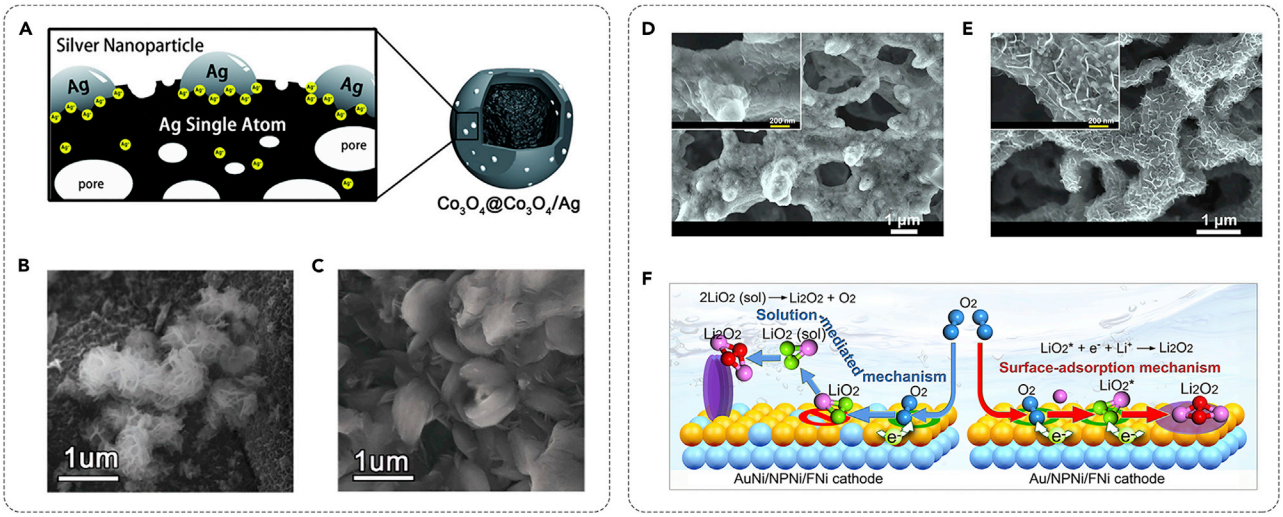


Figure 5. Design of Catalytic Materials in Non-aqueous Metal-Air Batteries

(A) Schematic illustration of Co_3O_4@Co_3O_4/Ag yolk-shell structure.

(B and C) SEM images of Co_3O_4@Co_3O_4/Ag-based (B) and Co_3O_4-based (C) air electrode after discharge. Reproduced with permission from Gao et al.[54] Copyright 2018, American Chemical Society.

(D and E) Field-emission SEM images of Au/NPNi/FNi (D) and AuNi/NPNi/FNi (E) air electrode after discharge with a limited discharge capacity of 3,000 mA h g^{-1}.

(F) A mechanism of Li_2O_2 electrochemical growth on AuNi/NPNi/FNi and Au/NPNi/FNi air electrode. Reproduced with permission from Xu et al.[56] Copyright 2017, American Chemical Society.

and part of the Ag is on the surface of Co_3O_4 shell in the form of single atoms or clusters. The Ag-Co_3O_4 interface tunes the electronic structure to provide active sites with higher activity for ORR and OER. When applied as the air electrode electrocatalyst of a Li-O_2 battery, Co_3O_4@Co_3O_4/Ag exhibited a high initial capacity (12,000 mA h g^{-1}@200 mA g^{-1}, based on the mass of electrocatalyst) and rate capability (4,700 mA h g^{-1}@800 mA g^{-1}). In a charge-discharge cycling test with a limited capacity of 1,000 mA h g^{-1} at the current density of 200 mA g^{-1}, the Li-O_2 battery catalyzed by Co_3O_4@Co_3O_4/Ag could retain the initial capacity after 80 cycles. This outstanding performance can be attributed to the highly active species, the electrical conductivity of Ag nanoparticles, the porous structure of Co_3O_4, and the modified structure of the discharge product Li_2O_2. Air electrodes with Co_3O_4@Co_3O_4/Ag and Co_3O_4 as electrocatalysts exhibited different morphology after discharge (Figures 5B and 5C). The flowerlike structure of Li_2O_2 on Co_3O_4@Co_3O_4/Ag possesses larger surface area, which is favorable for mass diffusion and Li_2O_2 decomposition. The change in Li_2O_2 morphology is benefited by the introduction of Ag, which provides nucleation sites for Li_2O_2 deposition.[55]

Although non-carbon electrocatalysts have been widely investigated, carbon materials are still utilized as catalyst supports or conductive additives in the air electrode of many Li-air batteries. Therefore it is necessary to develop non-carbon air electrodes, which are free of carbon in the whole electrode. This requires free-standing and conductive features of electrocatalysts. Considering this, Xu et al.[56] proposed an all-metal air electrode architecture of nanoporous Ni (NPNi) with a nanoengineered AuNi alloy surface directly attached to Ni foam (FNi), denoted as AuNi/NPNi/FNi. The Ni foam ensures the mechanical stability and the electrical conductivity of the air electrode; the porous Ni facilitates mass transfer; and the AuNi alloy provides ORR and OER activity. Therefore, the Li-O_2 battery with AuNi/NPNi/FNi as the air electrode exhibited ultrahigh capacity of 22,551 mA h g^{-1}_{AuNi},

and long cycling life (286 cycles) at a current density of 1.0 A g^{-1}_{AuNi} with a capacity limit of 3,000 mA h g^{-1}_{AuNi}. It is worth noting that the AuNi alloy can also regulate the morphology of discharge products. For the Li-O_2 battery with Au/NPNi/FNi as the air electrode, in which nanoporous Ni is coated by pure Au instead of AuNi alloy, the air electrode was covered by Li_2O_2 film after discharge (Figure 5D), while for AuNi/NPNi/FNi, Li_2O_2 showed a nanosheet-like structure (Figure 5E), similar to that in Figure 5B, which is favorable for achieving a high capacity. A possible mechanism is elucidated in Figure 5F. The Ni atoms in AuNi alloy prevent the reduction of LiO_2 on the surface and promote their diffusion into the electrolyte. The dissolved LiO_2 then undergoes disproportionation to form Li_2O_2 nanosheets. However, for AuNi/NPNi/FNi, the ORR occurs on the Au air electrode surface, resulting in the formation of Li_2O_2 thin films. As can be seen from the above research, regulation of the active material of electrocatalysts can accelerate the reaction kinetics, as well as tuning the structures of discharge products of non-aqueous metal-air batteries.

For a certain active material, a well-designed pore structure can bring about further improvements in the mass transport and the deposition of the discharge products. For conventional air electrodes with electrocatalysts coated or grown on conductive porous substrates, the specific capacity is limited by the surface passivation and low mass loading of the discharge products (Figures 6A and 6B).[57] Therefore, Lyu et al.[57] designed a 3D-printed hierarchically porous air electrode for Li-O_2 batteries, the schematic diagram of which is shown in Figure 6C. This hierarchical pore structure can provide sufficient space for Li_2O_2 storage, and the good contact between electrocatalysts and Li_2O_2 nanoparticles is favorable for Li_2O_2 decomposition. The air electrode design was then realized by the carbonization of 3D-printed Co-MOF, with Co nanoparticles assembled in nitrogen-doped nanoporous carbon flakes as active material, denoted as 3DP-NC-Co (Figure 6D). Compared with NC-Co randomly packed on carbon paper (RP-NC-Co) and NC-Co grown on carbon paper (CP-NC-Co), Li-O_2 batteries with 3DP-NC-Co as the air electrode exhibit much superior performances for specific capacity (Figure 6E), overpotential (Figure 6F), and cycle stability (Figure 6G). Therefore, electrocatalytic materials for non-aqueous metal-air batteries can be rationally designed from the aspects of active material and pore structure, and give rise to the cell performances.

The air electrode reactions of Na-air and K-air batteries are not exactly identical to those of Li-air batteries for the different discharge products: Na_2O_2 and NaO_2 for Na-air batteries, and KO_2 for K-air batteries.[4,9,58] As Na^+ and K^+ could more effectively stabilize superoxide ions than Li^+, lower ORR overpotentials can be obtained on Na-air and K-air batteries.[59] Nevertheless, material designs of Na-air and K-air batteries are still required to regulate the growth and decomposition mechanism of discharge products, thus achieving better discharge capacity and Coulombic efficiency.[60,61]

Gas-Diffusion Layer

In the air electrode of metal-air batteries, the GDL is outside the electrocatalyst layer. Primary roles of the GDL include ensuring efficient gas diffusion between the electrolyte and ambient air, preventing electrolyte flooding and leakage, and providing a mechanically stable and conductive support for electrocatalysts. The gas-diffusion layer for fuel cells has been well studied, and commercial GDLs have been developed, which can be directly employed in metal-air batteries. The most commonly used materials for GDLs are non-woven carbon paper and woven carbon cloth, both composed of carbon fibers. GDLs are often surface treated to reduce their affinity to the electrolyte, and coated by a microporous layer of carbon powder

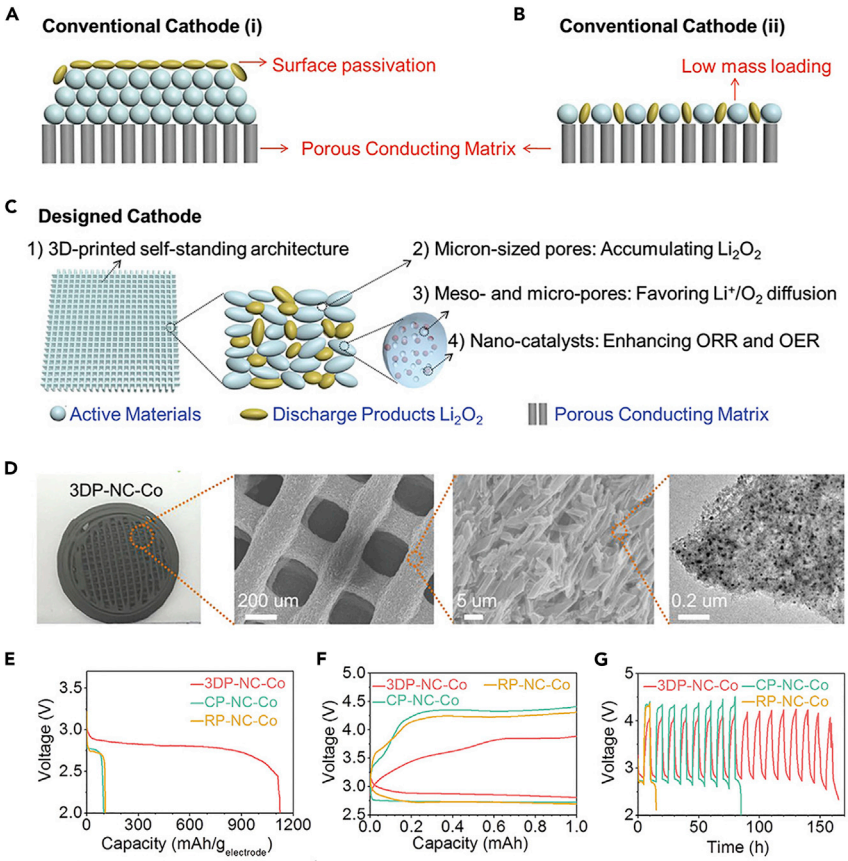


Figure 6. Design of Air Electrode Structure in Non-aqueous Metal-Air Batteries

(A and B) Schematic illustrations of Li_2O_2 growth on air electrodes with conventional structure.

(C) Schematic illustration and advantages of 3D-printed self-standing and hierarchically porous catalyst framework as the air electrode.

(D) Optical, SEM, and TEM images of 3DP-NC-Co.

(E–G) First discharge curves at the current density of 0.05 mA cm^{-2} (E), charge and discharge overpotentials at a limited capacity of 1 mA h (F), and cycle abilities with the current density of 0.1 mA cm^{-2} (G) of 3DP-NC-Co and contrast samples in Li-O_2 batteries. Reproduced with permission from Lyu et al.[57] Copyright 2019, Wiley-VCH Verlag GmbH & Co. KGaA.

on one side to strengthen the interaction with electrocatalysts. This type of commercial carbon fiber-based GDL has been widely utilized in metal-air batteries. Laboratory-made GDLs have also been reported. For example, Yang et al.[62] fabricated a polydimethylsiloxane (PDMS)-based flexible and stretchable GDL for use in flexible Li-O_2 batteries. Liquid PDMS was poured on Ni foam and filled in the holes, and after the solidification of PDMS, the removal of Ni foam template, and drying in vacuum, the GDL was obtained. This new type of GDL exhibits excellent mechanical stability and flexible ability, which are essential features of the flexible Li-O_2 battery.

As discussed in the design of electrocatalysts, carbon materials are easily corroded at high oxidation potentials. The corrosion of carbon-based GDLs may cause the degradation of the electrode, thus influencing the cycle life.[63] Therefore, metal-based GDLs with higher electrochemical stability and electrical conductivity have also attracted much attention. Metal-based materials such as Ni foam,[56,64] stainless-steel mesh,[63] and titanium mesh[65] have been investigated to serve as media for gas diffusion in both aqueous and non-aqueous metal-air batteries.

Up to now, commercial carbon papers or carbon cloths have been used as GDLs in most of the reported metal-air batteries. For modifications on GDLs and developments of new types of GDLs in future research, the following issues need to be addressed. (1) Commercial GDLs are mainly designed for aqueous fuel cells. Surface treatments such as a coating of polyvinylidene difluoride or polytetrafluoroethylene are applied to improve their hydrophobicity. However, such types of GDLs may not be suitable for organic electrolytes. Therefore, it is necessary to develop specialized GDLs according to the properties of electrolytes. (2) The electrocatalyst layers are usually combined with GDLs by drop casting or heat pressing. The weak interaction of electrocatalysts and GDLs may result in the immersion of electrocatalysts in the electrolyte, which blocks the contact between electrocatalysts and the oxygen. Therefore, the design of integrated air electrode structure is also worth investigating.

Material Design of Metal Electrodes

Metal electrodes of metal-air batteries face the challenges of metal dendrites, electrode shape change, corrosion, side reactions, and surface passivation.[10,66] For different types of metals, the key issues and the mitigation strategies can be different. In this section, the material design of the most investigated Zn and Li metal electrodes will be introduced, and advances in other metal electrodes will also be briefly presented. It is worth noting that the solutions to the problems of the metal electrode can also be obtained from the research on the electrolyte and the separator, which will be mentioned in subsequent sections.

Zn Electrode

Zinc-air batteries are usually filled with a high concentration of alkaline solution as the electrolyte. The reactions on the Zn electrode of alkaline Zn-air batteries are as follows:

$$Zn + 4OH^- \leftrightarrow Zn(OH)_4^{2-} + 2e^-, \quad \text{(Equation 5)}$$

$$Zn(OH)_4^{2-} \leftrightarrow ZnO + H_2O + 2OH^-. \quad \text{(Equation 6)}$$

In the discharge process, the metallic Zn is oxidized to Zn^{2+} and then converted to zincate in the strong alkaline solution. After the zincate ions are saturated, the ZnO will precipitate as the final discharge product. The deposition of insulating ZnO on Zn electrode leads to the passivation of Zn, which is a key factor that determines the specific discharge capacity of Zn-air batteries.[67] Moreover, the diffusion of zincate ions in the electrolyte makes the ZnO deposit not only on the Zn electrode but also on the air electrode and the inner side of the battery shell. Zinc oxides that are not deposited on the Zn electrode may cause the capacity loss and a low Coulombic efficiency. ZnO coated on the air electrode could also influence the performance of the oxygen electrocatalyst. Therefore, the main target of the material design for the discharge process in Zn-air batteries is to regulate the ZnO deposition.

Factors that may cause Zn electrode failure in the charge process can be classified into two categories. The first one is the morphology change. The concentration of zincates is unevenly distributed in the electrolyte, as a function of the distance from the Zn electrode surface. As a result, zincate ions are preferred to deposit on protrusions of the Zn electrode surface during charge.[10] With continuous cycling, sharp needle-like Zn dendrites would grow on the electrode surface, which can pierce the separator and even lead to short-circuit. Besides dendrites, changes on

the electrode thickness and surface area can also be observed during the cycling of Zn-based batteries.[68] The Zn electrode shape change can be attributed to the non-uniform distribution of the current and reaction zones on the electrode, bringing about densification of the electrode and a decrease in capacity.[34] The other issue is the hydrogen evolution on the Zn electrode. The hydrogen evolution reaction (HER) is the main side reaction on the Zn electrode of aqueous Zn-air batteries. The energy provided for charging would be partly consumed by hydrogen generation, resulting in low Coulombic efficiency. Moreover, the corrosion of Zn electrode could take place under non-operating status, due to the coupling of hydrogen evolution and zinc oxidation. This self-discharge process is also an important reason for the decay in capacity.

According to the above discussion, the main target for improving Zn electrode performance is to retain its original status after long discharge-charge cycles. Efforts made on the modification of Zn electrodes to improve their reversibility were focused on designing the electrode structure and introducing electrode additives.[10,15,34] Issues such as dendrites, ZnO deposition, and electrode corrosion are widespread problems for batteries with Zn as the metal electrode. Therefore, some design strategies for Zn electrode in other types of Zn-based batteries can also be applicable in Zn-air batteries.

An optimized structure of Zn electrode can control the dissolution/precipitation process of Zn and ZnO, and minimize the shape change of Zn electrode under long-time cycling.[69] Parker et al. proposed a 3D-structured Zn sponge with interconnected pores and long-range electrical conductivity, and have utilized it as the Zn electrode in various types of Zn-based aqueous batteries.[69–71] Figure 7A illustrates the application of 3D Zn sponge electrode in alkaline Ni-3D Zn batteries, in which the reactions on the Zn electrode are the same as those in Zn-air batteries.[71] In contrast to the severe dendrite problem on commercial Zn powder electrode, the 3D Zn sponge electrode exhibited minimal shape change with no dendrites observed. The high surface area and the monolithic structure of the 3D Zn sponge electrode also contributes to a high utilization efficiency of Zn. The battery achieved over 90% of the theoretical capacity based on the mass of Zn, and more than 95% of the capacity can be recovered in the charge process. The superior performance of the 3D Zn electrode can be attributed to the well-distributed current and electrolyte.[69] However, the reduced resistance of HER due to the high electrical conductivity and high surface area of the 3D-structured Zn electrode should also be fully considered.[34]

Introducing additives or surface coatings into Zn electrodes can regulate their chemical properties and interface structures. Electrode additives are expected to improve the Zn electrode performance from the following aspects: (1) regulating the surface structure, electrical conductivity, and current distribution to inhibit the growth of dendrites; (2) limiting the migration of $Zn(OH)_4^{2-}$ to minimize electrode shape change and irreversible consumption; and (3) stabilizing the Zn electrode and preventing electrode corrosion.

Carbon materials,[75,76] metals,[77–79] and metal compounds[72,80] have been investigated as additives for Zn electrodes to prevent dendrite formation. Kang et al.[72] coated nanoporous $CaCO_3$ on Zn foil to realize uniform Zn dissolution and precipitation. Figure 7B illustrates the mechanism for maintaining a smooth electrode surface by nano-$CaCO_3$ coating. On one hand, the high porosity of $CaCO_3$ layer induces uniform electrolyte flux and zinc dissolution/precipitation rate over the electrode surface. On the other hand, the insulating feature of $CaCO_3$ enlarges

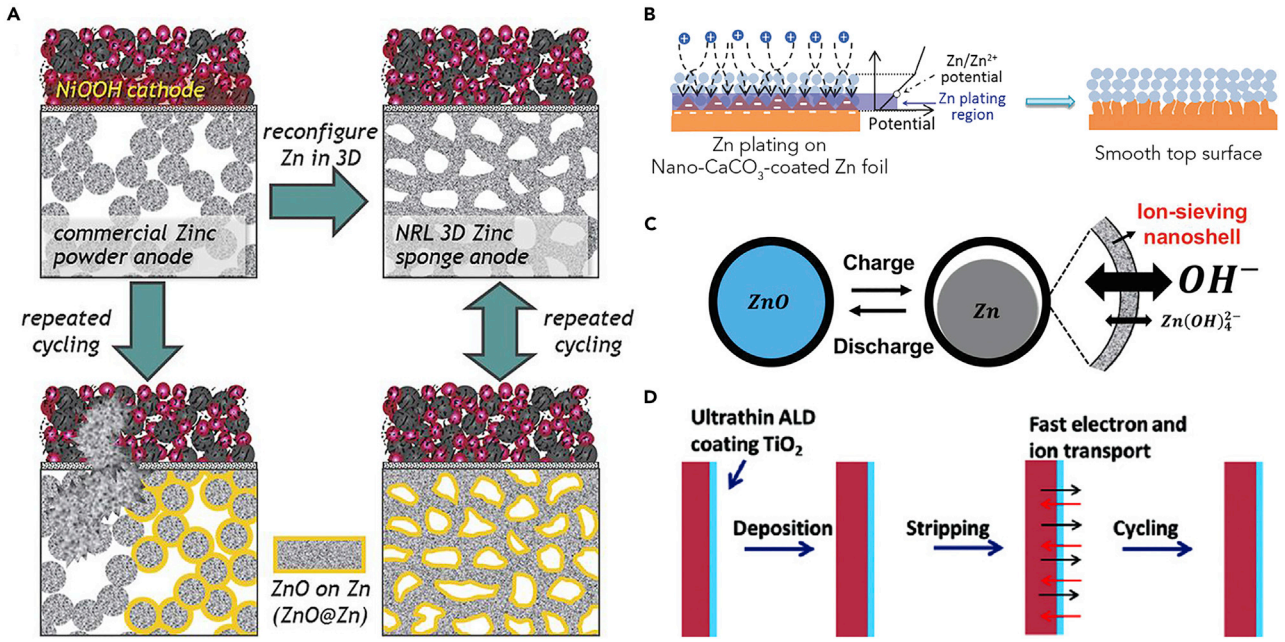


Figure 7. Regulation of the Structure and Composition of Zn Electrodes for Zn-Air Batteries

(A) Effect of the 3D sponge structure on the reversibility of Zn electrode. Reproduced with permission from Parker et al.[71] Copyright 2017, AAAS.

(B) Schematic illustrations of morphology evolution for nano-$CaCO_3$-coated Zn foils during Zn stripping/plating cycling. Reproduced with permission from Kang et al.[72] Copyright 2018, Wiley-VCH Verlag GmbH & Co. KGaA.

(C) The role of ion-sieving nanoshell on a ZnO nanoparticle on limiting zincate mitigation. Reproduced with permission from Wu et al.[73] Copyright 2018, Wiley-VCH Verlag GmbH & Co. KGaA.

(D) Schematic illustration of a thin TiO_2 to protect the Zn electrode. Reproduced with permission from Zhao et al.[74] Copyright 2018, Wiley-VCH Verlag GmbH & Co. KGaA.

the potential gradient across the $CaCO_3$ layer. As a result, only the area near the electrode surface can provide negative enough potentials for Zn formation in the charge process, thus preventing the accumulation of Zn on the protrusions. When applied in rechargeable aqueous batteries, Zn electrode with nano-$CaCO_3$ coating afforded 42.7% higher discharge capacity and over 15-fold longer cycle life than bare Zn foil.

The migration of $Zn(OH)_4^{2-}$ can be limited by ion-sieve coatings or additives with high affinity to $Zn(OH)_4^{2-}$. Carbon materials have been studied as Zn electrode coatings to limit zincate diffusion.[73,81] Wu et al.[73] synthesized carbon shell coated ZnO nanoparticles as the metal electrode of Zn-based aqueous batteries. As illustrated in Figure 7C, the carbon shell slows down the migration of zincate, and thus confines the anode shape change in a small space. Meanwhile, the carbon shell is permeable to smaller OH^- ions. When applied as metal electrode material in Ni-Zn batteries, the carbon shell coated ZnO exhibited about 1.6- and 6-fold longer cycle life than bare ZnO powder and Zn foil electrodes. Besides carbon materials, polymer coatings have also been used to control the zincate migration and the Zn electrode shape change.[82] For additives with high zincate affinity, Ca-based materials were widely investigated because they can form solid compound $Ca(OH)_2 \cdot 2Zn(OH)_2 \cdot H_2O$ with zincate. Carbon materials with oxygen-containing groups have also been reported to form a hydrogen bond with zincates.[81]

The stabilization of Zn electrode can be achieved by adding metals,[83,84] metal compounds,[74,85] and polymers[86] in the electrode. Recently, ultrathin TiO_2 coating on Zn

plate was reported by Zhao et al. to mitigate hydrogen evolution and Zn corrosion in alkaline Zn-MnO_2 batteries.[74] The ultrathin TiO_2 of ~5 nm was coated via an atomic layer deposition process. As illustrated by Figure 7D, the ultrathin TiO_2 layer avoids the direct contact of the Zn electrode and the electrolyte while allowing fast electron and ion transport. The Zn-MnO_2 battery with the TiO_2-coated Zn plate as the metal electrode can maintain 85% of the initial capacity after 1,000 cycles, exhibiting much higher stability than pristine Zn plate with a cycle life of about 400 cycles.

Given the above, regulations on the structure and composition of Zn electrodes can mitigate the problems of dendrites, shape change, passivation, and corrosion, leading to a better reversibility of Zn electrodes. Similar strategies have also been applied on Al and Mg electrodes in aqueous batteries with similar challenges. For example, Ma et al.[87,88] synthesized a series of Al-based alloys as the metal electrode of Al-air batteries. The utilization efficiency and self-corrosion resistance of the metal electrode were improved through the introduction of metal alloy electrodes.

Li Electrode

Li metal electrode is considered the "Holy Grail" in energy-storage systems because of its low density (0.59 g cm^{-3}), high theoretical specific capacity (3,860 mA h g^{-1}), and the lowest negative redox potential (−3.04 versus standard hydrogen electrode).[89] Lithium metal batteries (LMBs), including Li-S and Li-air batteries, have been extensively studied as next-generation energy-storage devices.[90,91] In the research on LMBs, the inhibition of Li dendrites has attracted most attention.[92] Similar to Zn dendrites as discussed above, Li dendrites can lead to degradations in specific capacity, Coulombic efficiency, stability, and so forth. More seriously, as Li is much more active than Zn, the piercing of the separator and the circuit shortcut can cause severe safety issues.[93] In addition to the dendrite problem, which is universal for all types of LMBs, the Li electrode in Li-air batteries also suffers from the O_2 crossover effect.[94] The diffusion of O_2 from the air electrode induces the electrolyte decomposition and the formation of LiOH and carbonates on the Li electrode.[95] Meanwhile, the influence of H_2O, CO_2, and N_2 on the Li electrode stability when Li-air batteries are operated in ambient air should not be ignored.[96] To overcome these problems, material design methods for Li electrode, electrolyte, and separator have been studied.[6] In this section, modification strategies for Li metal electrode are outlined.

The performance of Li metal electrodes can be improved by applying a 3D Li host,[97–100] tuning the electrode composition[19,101–103] and surface coating.[101,104–107] One important target of applying these regulation methods is to control the current distribution and ion flow, thus changing the nucleation and deposition behavior of Li and inhibiting the Li dendrite growth.[99,108] Recently, Guo et al.[109] applied this strategy to stabilize Li metal anodes for Li-air batteries by using lithiophilic Co/Co_4N nanoparticles to regulate lithium nucleation and deposition. The anode was fabricated by depositing Li on MOF-derived hollow N-doped carbon-based cubes with uniformly embedded Co/Co_4N nanoparticles (Co/Co_4N-NC), as illustrated in Figure 8A. Lithiophilic functional groups such as N heteroatoms in carbon and Co_4N promote the homogeneous nucleation of Li deposits, and the porous carbon structure provides high Li-storage capacity. Therefore, the lithiated Co/Co_4N-NC electrode exhibits a smooth and dendrite-free morphology (Figure 8B). When serving as the metal electrode of a Li-air battery, the lithiated Co/Co_4N-NC electrode remained reproducible after 70 cycles at a current density of 200 mA g^{-1} and a limited specific capacity of 500 mA h g^{-1} (Figure 8C). In contrast, pure Li electrode failed after only 34 cycles in the same testing condition.

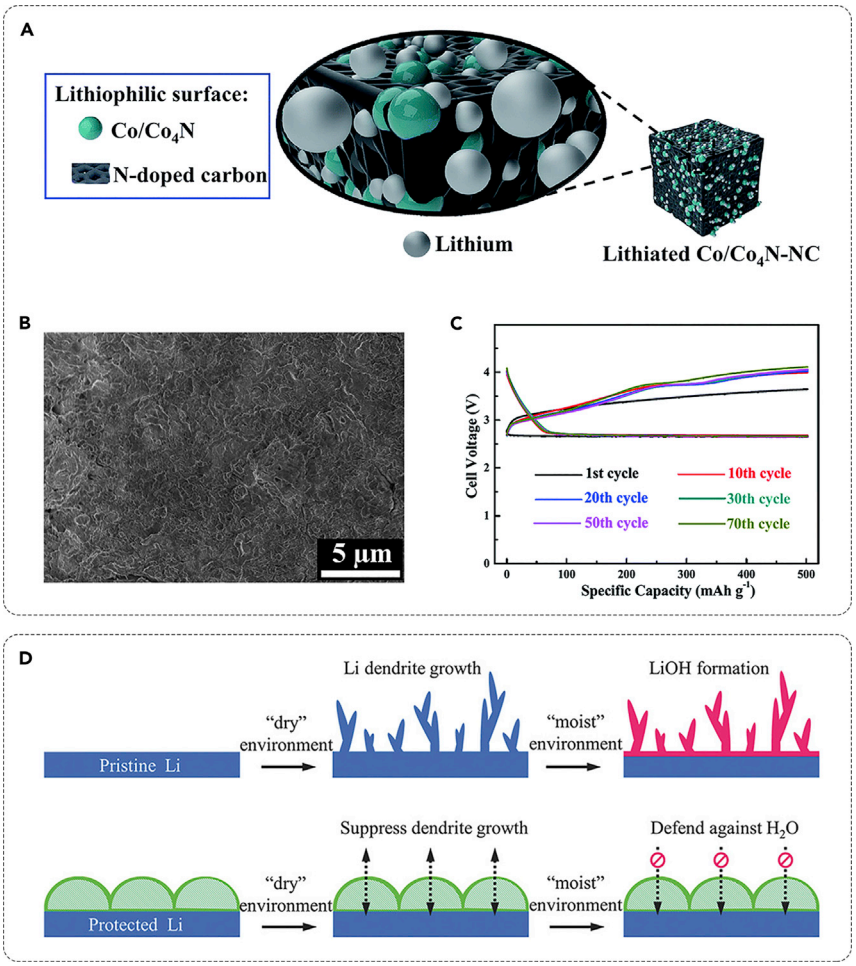


Figure 8. Design of Lithium-Containing Electrodes and Lithium Surface Coatings for Li-Air Batteries

(A) Schematic illustration of lithiated Co/Co_4N-NC nanoparticles.

(B) SEM image of Co/Co_4N-NC electrode after Li deposition with a limited capacity of 2.0 mA h cm^{-2}.

(C) Discharge/charge profiles of the Li-air battery using the lithiated Co/Co_4N-NC electrode at a current density of 200 mA g^{-1} with a limited capacity of 500 mA h g^{-1}. Reproduced with permission from Guo et al.[109] Copyright 2018, Royal Society of Chemistry.

(D) Schematic illustration of Li plating/stripping on pristine lithium electrode and protected lithium electrode. Reproduced with permission from Liao et al.[105] Copyright 2018, Wiley-VCH Verlag GmbH & Co. KGaA.

The solid electrolyte interface (SEI) formed on the surface of Li electrode in Li metal batteries is considered to be favorable for enhancing the cycling ability of the Li electrode as a protective layer.[101,110] Therefore, approaches have been developed to improve the stability and uniformity of SEI, or build an artificial SEI to stabilize Li electrode.[90,104–107] The protective layer for the Li electrode in Li-air batteries is required to afford properties such as high mechanical strength to suppress the Li dendrite growth, low O_2 and H_2O permeability to protect the Li electrode, good electronic insulation to avoid electrolyte decomposition, high ionic conductivity to ensure fast Li-ion diffusion, and high stability. Considering these requirements, Liao et al.[105] constructed a Ge-based artificial SEI layer on the Li surface to allow stable cycling of Li-O_2 batteries. The protective layer was generated by immersing Li foil in $GeCl_4$-tetrahydrofuran steam. The as-synthesized film is about 1.5 μm thick with

micrometer-scale spheres containing amorphous germanium (Ge and GeO_x) and lithium compounds (Li_2CO_3, LiOH, Li_2O, and LiCl). This protective layer is robust enough to suppress the Li dendrite growth and keep the Li surface smooth after Li plating/stripping cycling, and is also able to protect the Li electrode against H_2O attack, as shown in Figure 8D. As a result, the Li-O_2 battery with Ge-based artificial SEI layer on Li electrode can be stably discharged and charged over 150 cycles in humid O_2 with 45% relative humidity.

Na and K electrodes face challenges similar to those of the Li electrode, including dendrite growth and O_2 crossover. Similar strategies have also been developed to improve the performances of Na and K electrodes. For example, McCulloch et al.[111] demonstrated the role of K-Sb alloy in preventing dendrite growth and O_2 crossover in K-O_2 batteries. Chi et al.[112] used a 3D flexible carbon felt host for Na deposition, and achieved stable dendrite-free cycling in Na metal batteries.

The development of stable and reversible metal electrodes for metal batteries has attracted much attention in recent years, and most of the battery performances are tested in a closed system.[92,113] For the open system of metal-air batteries, the side reactions on the metal electrode can be more complicated.[114] Therefore, specific requirements for metal-air batteries need to be considered when adopting the metal electrode design methods for other metal-based batteries. More comprehensive methodologies for metal-air battery metal electrode regulation also need to be established.

Material Design of Electrolytes

Aqueous Electrolytes

Metal-air batteries with water-stable metal electrodes, such as Zn-air batteries and Al-air batteries, usually use aqueous alkaline electrolytes.[5,8] Compared with acidic and neutral electrolytes, metal-air batteries with alkaline electrolytes can afford higher metal corrosion resistance and better oxygen electrocatalysis activity.[34] Among the commonly used alkaline electrolytes including KOH, NaOH, and LiOH, concentrated KOH solutions, with the highest ionic conductivity and the lowest viscosity, are most frequently employed in metal-air batteries.[34]

Alkaline electrolytes are sensitive to the CO_2, resulting in the formation of carbonates and decrease in electrolyte concentration. The adsorption of CO_2 has been confirmed to have negative effects on the capacity and cycle life of Zn-air batteries.[5] Moreover, strong alkaline electrolytes are also environmentally unfriendly. Therefore, proper neutral electrolytes for metal-air batteries have been studied.[115,116] For the future development of neutral electrolytes, electrocatalysts with high ORR and OER activity in neutral conditions are highly desirable.

As already mentioned, metal electrodes in aqueous metal-air batteries suffer from issues such as dendrite growth, passivation, and hydrogen evolution. Some of these issues can be mitigated by the material design of the electrolyte.[77,117] In addition, aqueous electrolytes have also been used in the air electrode side of Li-air batteries to enhance the tolerance to ambient air and the electrolyte stability.[118] In this case, solid-state glass ceramic electrolytes are usually used to protect the Li electrode.[5,29]

Aqueous Electrolytes for Improving Metal Electrode Performance. Much research on Zn-air batteries used 6 M KOH solution with 0.2 M zinc chloride or zinc acetate as the electrolyte.[7,31] The additional zinc ions convert to zincate ions in strong alkaline solution, which can improve the reversibility of the metal electrode and the

rechargeability of the battery.[119] Zincate ions generated on the Zn electrode rapidly precipitate as ZnO in a concentrated zincate solution and thus the migration of zincate ions is limited, leading to a minimized shape change. Besides the introduction of zincate ions, other strategies have also been applied on the aqueous electrolyte to achieve high reversibility of metal electrodes.

As introduced above, the reversibility of Zn electrodes can be enhanced by limiting the diffusion of zincate ions. Modifications on the KOH electrolyte such as reducing the KOH concentration and adding potassium salts have been investigated to reduce the zincate solubility, thus minimizing the Zn electrode shape change.[34] Up to now, this kind of strategy was mainly applied on other types of Zn-based aqueous batteries such as Ni-Zn batteries.[120] Investigations on limiting zincate migration through electrolyte additives in Zn-air batteries are promising for future research.

The reversibility of Zn electrodes can also be enhanced by changing the existing form of zinc species in the electrolyte. Wang et al.[28] proposed a highly concentrated Zn-ion electrolyte (HCZE) to realize dendrite-free Zn plating/stripping at nearly 100% Coulombic efficiency. In this 1 mol kg^{-1} $Zn(TFSI)_2$ + 20 mol kg^{-1} LiTFSI electrolyte (TFSI, bis(trifluoromethanesulfonyl)imide), zinc ions are surrounded by TFSI ions instead of water. The Zn^{2+}-solvation sheath occupied by TFSI suppressed the hydrogen evolution on the Zn surface, leading to a reversible and dendrite-free Zn plating/stripping. In a Pt/Zn coin cell with HCZE, the Zn plating and stripping at a constant current density exhibited ~99.5% Coulombic efficiency after the first three cycles (Figure 9A). When applying HCZE in a Zn-O_2 cell with carbon paper as air electrode, the cell delivered a capacity of ~3,000 mA h g^{-1} and discharge/charge voltages of around 0.9/1.9 V at 50 mA g^{-1} (Figure 9B). The Zn-O_2 battery also showed good stability after 200 cycles at a limited capacity of 1,000 mA h g^{-1} and a current density of 50 mA g^{-1} (Figure 9C). Lee et al.[121] used alkoxide and acetate ions as additives in a KOH electrolyte to replace hydroxide ions in zincate. The formation of ZnO was suppressed by this method, resulting in enhanced reversibility of the Zn electrode.

Electrolyte additives have also been utilized to enhance the performance of metal electrodes beyond zinc. Aluminum is highly active in alkaline solutions, and electrolyte additives, named as corrosion inhibitors in aluminum electrode research, have been investigated to prevent Al self-corrosion in Al-air batteries.[8,122] For example, Liu et al.[123] found that, with a small amount of carboxymethyl cellulose (CMC) and ZnO in 4 M NaOH electrolyte, the corrosion of aluminum electrode in Al-air batteries can be effectively mitigated, which is due to the protective layer of CMC and Zn-ion complex on the electrode surface. Flax straw extract containing mainly humic and fulvic acids was also demonstrated to be effective in suppressing Al corrosion, as reported by Grishina et al.[124] Aluminum ions can react with carboxylic acid and amino groups in such natural corrosion inhibitors, thus forming a protective layer on the Al electrode surface.

Aqueous Electrolytes for Li-Air Batteries. The use of aqueous lithium salt solutions as electrolytes for Li-air batteries can avoid the side reactions and electrolyte decomposition problems caused by organic solvents. However, the application of aqueous electrolytes for Li-based batteries is limited by their narrow potential window.[125] The high-voltage operation of aqueous electrolytes can be achieved by eliminating free water molecules in the electrolyte, for example by using high concentration water-in-salt (WiS) electrolytes[126] or hydrate-melt electrolytes.[127] Recently, Dong et al.[27]

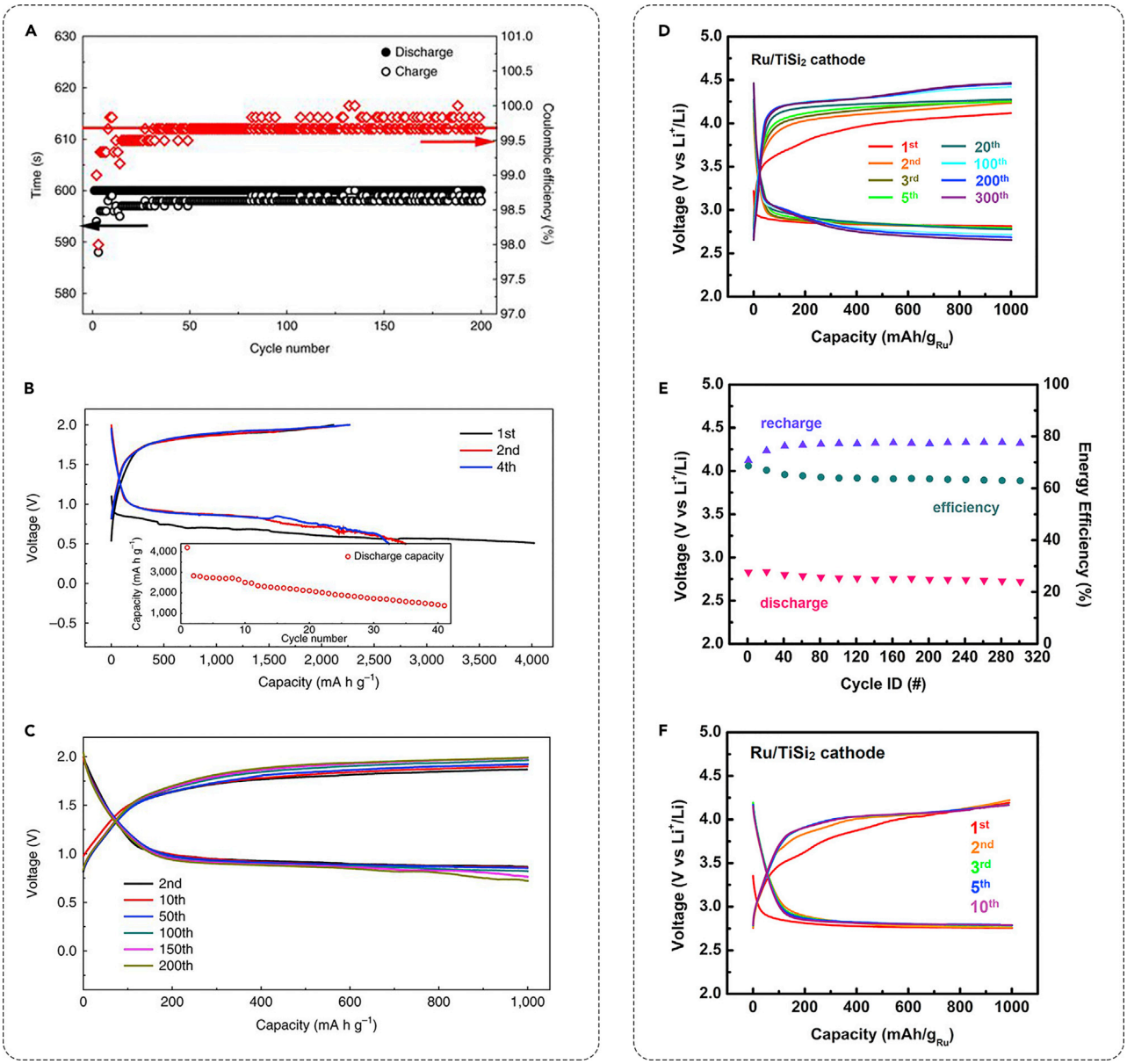


Figure 9. Modifications of Aqueous Electrolytes to Improve Metal Electrode Performance to Be Employed in Li-Air Batteries

(A) Zn plating/stripping time (left) and Coulombic efficiency at the current density of 1 mA cm^{-2} on Pt electrode in HCZE.

(B) Full-range voltage profile of the Zn-O_2 battery in HCZE with 70 wt % super P as the air electrode at a constant current of 50 mA g^{-1} (based on the air electrode).

(C) Cycling performance of the Zn-O_2 battery in HCZE at a constant current of 50 mA g^{-1} and a limited capacity of 1,000 mA h g^{-1}. Reproduced with permission from Wang et al.[28] Copyright 2018, Springer Nature.

(D) Cycling performance of the Li-O_2 battery in WiS electrolyte with LFP as pseudo-metal electrode and Ru/$TiSi_2$ as air electrode.

(E) Average voltage and round-trip energy efficiencies as a function of cycling numbers calculated from the data in (D).

(F) Cycling performance of the Li-O_2 battery in WiS electrolyte with protected Li metal electrode and Ru/$TiSi_2$ air electrode. Reproduced with permission from Dong et al.[27] Copyright 2018, Elsevier.

designed a water-in-salt system with superconcentrated 21 mol kg^{-1} LiTFSI to serve as the electrolyte for Li-O_2 batteries. The Li-O_2 battery with WiS electrolyte, Ru/$TiSi_2$ as air electrode, and $LiFePO_4$ (LFP) as pseudo-metal electrode exhibited stable cycling performance for 300 cycles at a discharge depth of 1,000 mA h g^{-1}_{Ru} (Figure 9D), and the energy efficiency remained higher than 62% (Figure 9E).

When using glass ceramic-protected Li as metal electrode, the Li-O_2 battery delivered good polarization performance but poor stability (10 cycles, Figure 9F). Therefore, the engineering of the electrolyte and the electrode should be considered together in the full cell design of metal-air batteries.

Organic Electrolytes

Aprotic organic electrolytes are widely employed in metal-air batteries with metal electrodes that are unstable in water, such as Li and Na. Optimizations of this type of electrolyte have been studied with respect to the organic solvents, metal salt solutes, and electrolyte additives. The material designs are mainly aimed at improving the stability of the electrolytes and enhancing the air/metal electrode performance.

Electrolyte Stability. In metal-air batteries, the superoxide species generated on the air electrode can cause the decomposition of organic solvents, forming CO_2, H_2O, and metal carbonates.[33,45] Therefore, unceasing efforts have been made in searching for stable organic solvents that meet the requirements for metal-air batteries, including good metal ion conductivity, wide potential window, high oxygen solubility, low volatility in air, and high solvation of superoxides/peroxides.[46] Carbonate-based solvents were firstly investigated but were then found to be unstable under the attack of superoxide species.[33,45] Ether-based solvents were then studied, and confirmed to be more insusceptible to reduced oxygen species than carbonate-based solvents.[128] However, research on Li-air batteries has revealed that the O_2 crossover could induce the decomposition of ether-based electrolytes on the Li electrode.[95] Other types of organic solvents including sulfoxide-based and amide-based solvents have also been employed in metal-air batteries.[58] For these solvents, importance should also be attached to their stability on metal electrodes. A protective layer on the metal electrode is usually needed for better electrolyte stability.[46,58] Theoretical studies of the mechanism of electrolyte decomposition are also expected to facilitate the future development of new organic electrolyte solvents.

Metal salt solutes are another essential part of organic electrolytes. Highly soluble metal salts with stable anions are recommended for metal-air batteries. Taking Li-air batteries as an example, lithium salts including $LiPF_6$, $LiClO_4$, LiBr, $LiNO_3$, LiTFSI, and $LiCF_3SO_3$ have been studied.[129] The discharge product Li_2O_2 and other oxygen species are the major cause of the decomposition of lithium salts, especially for that with F-containing anions.[6] The stability of metal salts also varies with the type of electrolyte solvents and the concentration.[130,131] Therefore, the optimization of organic solvents and metal salts should be considered together for better electrolyte stability.

Organic Electrolytes for Improving Air Electrode Performance. Regulation of the electrolyte in non-aqueous metal-air batteries can improve the air electrode performance by increasing the superoxide/peroxide storage during the discharge process and facilitating the superoxide/peroxide decomposition during charge process. Influences of electrolyte solvents, metal salts, and additives on the Li_2O_2 generation and decomposition have been intensively studied in Li-O_2 batteries.[46,132]

On the air electrode of Li-O_2 batteries, oxygen is first reduced to O_2^-. The solubility of superoxide intermediates depends on the donor number (DN) of the electrolyte solvent, which represents the solvent basicity according to the HSAB theory.[46] Solvents with high DN can afford higher stability for the (Li^+-[solvent]$_n$-O_2^-) complex[29,45] and induce a solution growth mechanism for Li_2O_2 formation, whereby soluble LiO_2 disproportionates into Li_2O_2 and O_2 in solution. In contrast, the oxygen

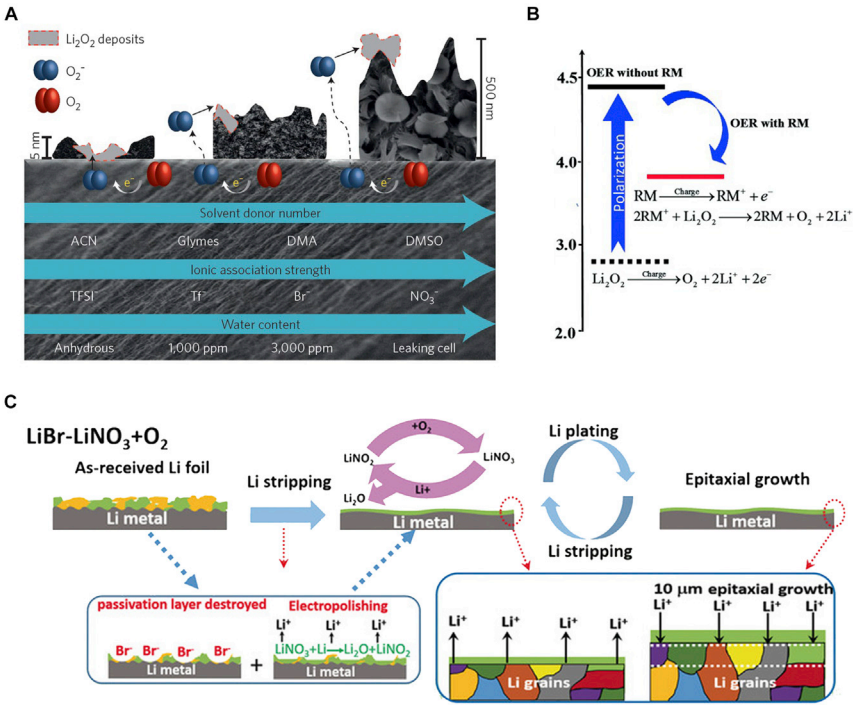


Figure 10. Material Designs of Non-aqueous Electrolytes with Organic Solvents and Their Effects on the Performance of Air Electrode and Metal Electrode of Metal-Air Batteries

(A) Effects of solvents, lithium salts, and water content on the Li_2O_2 growth mechanism on air electrodes. can, acetonitrile; glymes, polyethers ($CH_3O(CH_2CH_2O)_nCH_3$); DMA, dimethylacetamide; DMSO, dimethyl sulfoxide; Tf^-, trifluoromethanesulfonate. Reproduced with permission from Aurbach et al.[133] Copyright 2016, Springer Nature.

(B) Diagram of the mechanism and the suitable redox potential region of redox mediators. Reproduced with permission from Feng et al.[134] Copyright 2015, Wiley-VCH Verlag GmbH & Co. KGaA.

(C) Schematic illustration of dendrite-free epitaxial growth of Li in $LiBr\text{-}LiNO_3$ electrolyte under O_2 atmosphere. Reproduced with permission from Xin et al.[135] Copyright 2018, Wiley-VCH Verlag GmbH & Co. KGaA.

reduction in low-DN solvents obeys the surface mechanism. LiO_2 is adsorbed on the electrode surface and then electrochemically reduced to an insulating film of Li_2O_2 on the Li surface.[132] As a result, electrolytes with high-DN solvents are beneficial for achieving high capacity in metal-air batteries. In addition, the lithium salts and water content in the electrolyte also affect the Li_2O_2 growth mechanism, as shown in Figure 10A.[133] This indicates that a trace amount of water in the electrolyte can be favorable for the air electrode performance, while the parasitic reactions induced by water should also not be ignored.

In the charge process of Li-O_2 batteries, the oxidation of Li_2O_2 primarily occurs at the contacting part of the Li_2O_2 and the solid electrocatalyst, resulting in insufficient charging and low Coulombic efficiency. Therefore, a specific type of electrolyte additive, called redox mediators (RMs), has been employed in Li-O_2 batteries to promote the Li_2O_2 decomposition by catalyzing the reaction in the solution.[23] During charging, RMs are first oxidized on the air electrode, after which the oxidized RMs react with Li_2O_2 to form Li^+ and O_2. A suitable RM for this process needs to be soluble and stable in the electrolyte, and its redox potential should be in an appropriate range to reduce the charge overpotential, as illustrated in Figure 10B.[134] Up to now various materials including organic, organometallic, and halide materials have been investigated as RMs in Li-O_2 batteries.[32] RMs for the ORR process have also been

developed to regulate the Li_2O_2 formation and improve the discharge capacity.[136] This idea of transforming solid-solid reactions to liquid-solid reactions is also promising for promoting the reduction of ZnO in Zn-air batteries.

Organic Electrolytes for Improving Metal Electrode Performance. One strategy in the material design of the metal electrode is to build a protective layer on the metal electrode to suppress the dendrite growth and avoid side reactions.[101] The protective layer can be coated on the metal electrode before the fabrication of a metal-air battery, and can also be generated in the metal-air battery with the assistance of the electrolyte.[135,137,138] The formation of an ultrathin and homogeneous SEI layer on the Li electrode in a Li-O_2 battery with LiBr-$LiNO_3$ dual-anion electrolyte was reported by Xin et al.[135] Figure 10C schematically illustrates Li stripping and plating in the LiBr-$LiNO_3$ electrolyte. In the discharge process, Br^- ions promote the corrosion of the original irregularly shaped SEI layer, and NO_3^- ions immediately passivate the Li surface to form a thin Li_2O-rich SEI layer. This thin and homogeneous SEI layer contributes to a dendrite-free epitaxial growth of Li in the charge process. As a result, a stable cycle performance of more than 900 h at the current density of 0.5 mA cm^{-2} was achieved in a symmetrical Li/Li cell with LiBr-$LiNO_3$ electrolyte.

Ionic Liquid Electrolytes

Room-temperature ionic liquids are a type of salts whose melting point is below 100°C and composed of large organic cations and organic/inorganic anions. Ionic liquids are promising alternatives for the electrolyte solvents in both aqueous and non-aqueous electrolytes because of their wide potential window, low volatility, high ionic conductivity, and nonflammability.[6,16,34] Ionic liquids can benefit metal-air battery performance from various aspects. Shiga et al.[139] employed ionic liquid electrolyte in a Ca-O_2 battery to enhance the reversibility of Ca stripping/plating. Thomas et al.[140] designed a Li-air battery with [Li(G4)][TFSA] electrolyte (G4 = tetraglyme, TFSA = bis(trifluoromethanesulfonyl)amide). Benefited by the hydrophobic nature of ionic liquids, the Li-air battery exhibited high moisture resistance. Ionic liquids were also studied as electrolyte additives. A tetramethylpiperidinyloxyl (TEMPO)-grafted ionic liquid (IL-TEMPO) was synthesized by Zhang et al.[141] and employed as an electrolyte additive in Li-O_2 batteries. The TEMPO group functions as the redox mediator and oxygen shuttle, and the imidazolium moiety facilitates the formation of a stable SEI layer. The IL-TEMPO can also serve as the electrolyte solvent to enable the operation in atmospheric air. Asadi et al.[142] reported a Li-air battery with a mixed electrolyte of 25% ionic liquid 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIM-BF4) and 75% dimethyl sulfoxide (DMSO). This electrolyte is stable in the Li-air system, and affords high LiO_2 solubility to ensure the solution growth mechanism of Li_2O_2. The strong binding of H_2O and CO_2 in the ionic liquid also prevents side reactions on Li_2O_2. Together with material designs of the protected Li electrode and the air electrode electrocatalyst, a lithium-oxygen battery with a long cycle life in an air-like atmosphere was demonstrated.

Solid-State Electrolytes

Compared with liquid electrolytes, solid-state electrolytes can avoid the problems of electrolyte evaporation, electrolyte leakage, and gas crossover.[6] Moreover, the application of solid-state electrolytes is favorable for improving the energy density of metal-air batteries, and also enables the fabrication of flexible and wearable devices.[143] For the development of solid-state electrolytes for metal-air batteries, suitable materials with high ionic conductivity, wide potential window, and chemical stability are required, and the insufficient solid-solid interfacial contact with the electrolyte and electrodes should also receive much attention.[144]

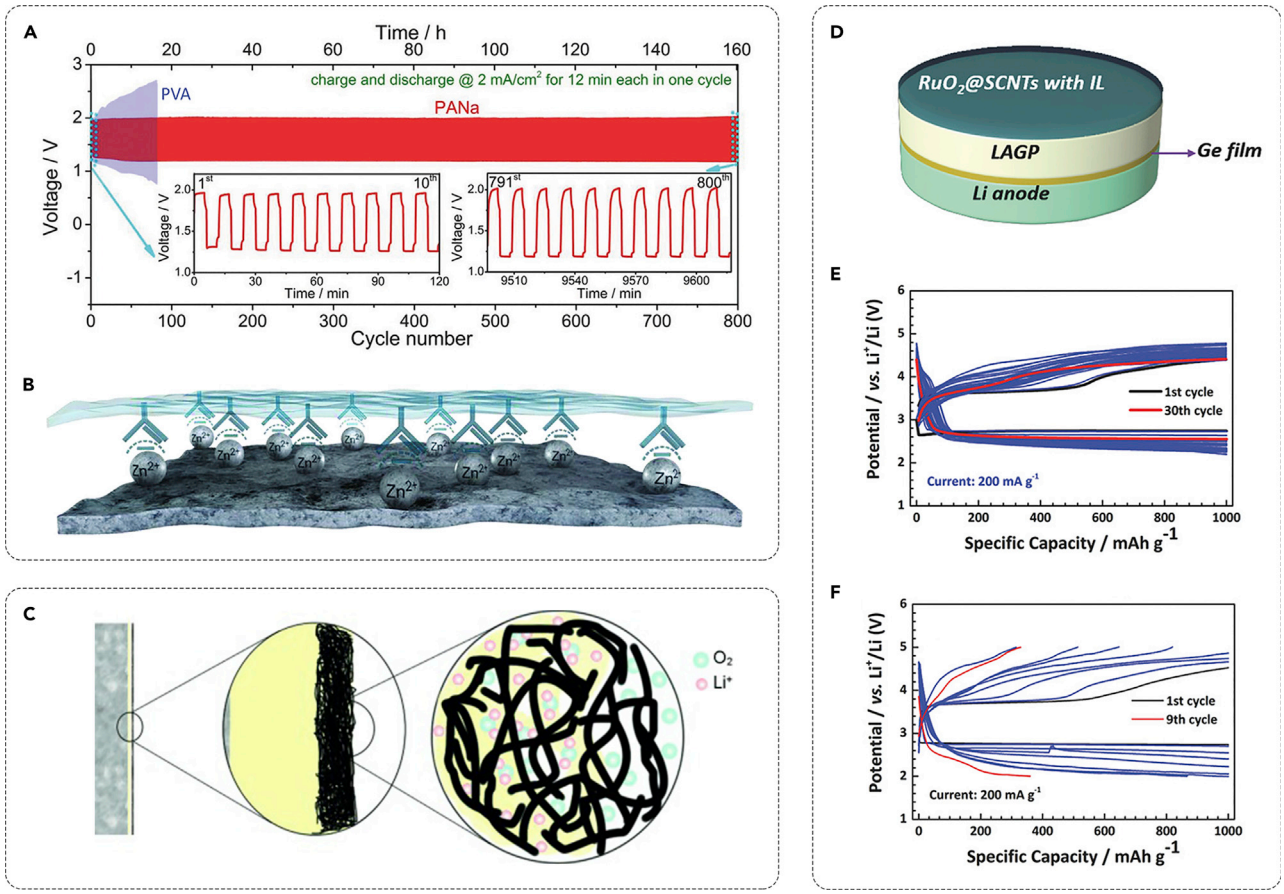


Figure 11. Applications and Optimization Methods of Solid-State Electrolytes for Metal-Air Batteries

(A) Galvanostatic discharge-charge cycling curves at 2 mA cm^{-2} of Zn-air batteries with PANa and PVA gel polymer electrolytes.

(B) Schematic illustration of the formation of quasi-SEI to prevent Zn dendrite growth. Reproduced with permission from Huang et al.[147] Copyright 2018, Wiley-VCH Verlag GmbH & Co. KGaA.

(C) Schematic illustration of a Li-O_2 battery with a solid polymer electrolyte. Reproduced with permission from Balaish et al.[150] Copyright 2015, Wiley-VCH Verlag GmbH & Co. KGaA.

(D) Schematic diagram of a Li-air battery with Ge film-coated LAGP solid-state electrolyte.

(E and F) Cycling performance of Li-O_2 batteries with Ge film-coated LAGP (E) and uncoated LAGP (F) at a constant current of 2,000 mA g^{-1} and a limited capacity of 1,000 mA h g^{-1}. Reproduced with permission from Liu et al.[151] Copyright 2018, Wiley-VCH Verlag GmbH & Co. KGaA.

Gel-Polymer Electrolytes. Gel-polymer electrolytes consist of a 3D polymer skeleton with a liquid electrolyte expanded throughout its whole volume to afford fast ion transport similar to liquid electrolytes, as well as high flexibility. In recent years gel-polymer electrolytes have been widely investigated in both aqueous and non-aqueous flexible metal-air batteries.[26,145,146]

Metal-air batteries with gel-polymer electrolytes suffer from the limited stability of polymer frameworks and the quick loss of electrolyte solvents, resulting in short cycle life. Therefore, stable, solvent-retaining gel-polymer electrolytes are highly desired for extending the battery lifetime. Considering this, Huang et al.[147] designed a sodium polyacrylate hydrogel (PANa) for flexible Zn-air batteries. PANa is well known as a superabsorbent material with the ability to absorb water as much as 100 to 1,000 times its mass, and can keep its mechanical properties even when saturated with highly concentrated solutions. Therefore, the Zn-air battery with PANa electrolyte exhibited stable cycling performance at 2 mA cm^{-2} for 160 h (Figure 11A), significantly superior to traditional polyvinyl alcohol (PVA)

electrolyte, which showed a rapid drop of round-trip efficiency from 60.4% to 29.3% after only 16 h. Moreover, the electrostatic interactions between acrylate ions in PANa and Zn ions induce the formation of a quasi-SEI layer to avoid Zn dendrite growth (Figure 11B). For flexible Li-air batteries, gel-polymer electrolytes have been reported to ensure the flexibility and protect the Li electrode from corrosion in ambient air.[148,149]

Solid Polymer Electrolytes. Similar to gel-polymer electrolyte, a solid polymer electrolyte is composed of an ion-conducting polymer matrix, but not swollen with liquid solvents. Due to the absence of liquid solvents, the ionic conductivity in solid polymer electrolytes is limited.[152] Commonly investigated polymer materials for solid polymer electrolytes include polyacrylonitrile (PAN), poly(carbonate-ether), and poly(ethylene oxide) (PEO).[6] Figure 11C schematically illustrates a Li-O_2 battery with PEO-based electrolyte proposed by Balaish et al.[150] However, this Li-O_2 battery was operated at 80°C, above the melting point of the poly(ethylene oxide)-lithium triflate electrolyte, to obtain acceptable ionic conductivity. Theoretical studies on the ion-conduction mechanisms revealed that polymers with lower glass transition temperature (T_g) and more amorphous structure could provide higher ionic conductivity.[152] Therefore, material design methods such as crosslinking and synthesizing copolymers have been proposed to reduce the T_g and achieve high ionic conductivity.[152]

Ceramic Electrolytes. Inorganic ceramic electrolytes, including LISICON (Li^+ superionic conductor), NASICON (Na^+ superionic conductor), perovskite, garnet, and sulfides, have demonstrated their applications in solid-state Li and Na metal batteries.[144] The high mechanical strength of ceramic electrolytes is favorable for mitigating the dendrite growth on the metal electrode.[152] For better performance of metal-air batteries with ceramic electrolytes, the material design on ceramic electrolytes mainly focuses on the following points: the ionic conductivity, the stability with Li/Na metal, and the electrolyte-electrode interfacial contact.[143,144,151,153]

Liu et al.[151] introduced a protective film between the Li electrode and a NASICON type solid-state electrolyte to protect the electrolyte and enhance its contact with the Li electrode. The NASICON type solid electrolyte LAGP ($Li_{1.5}Al_{0.5}Ge_{0.5}P_3O_{12}$) exhibits high lithium ionic conductivity and good mechanism strength, but Ge^{4+} ions are feasible to be reduced to Ge^0 and Ge^{2+} on the Li electrode, leading to a decrease on the Li-ion conductivity. To alleviate this problem, an amorphous Ge thin film is introduced between LAGP and the Li electrode, as illustrated in Figure 11D. This Ge film can suppress the reduction of Ge^{4+} and also form a Li ionic interlayer to ensure intimate contact between Li metal and the solid electrolyte. As a result, the solid-state Li-air battery with Ge/LAGP electrolyte exhibited stable cycling performance at 200 mA g^{-1} for 30 cycles (Figure 11E), while the LAGP electrolyte without Ge coating showed much inferior stability (Figure 11F). From the above discussion, improving the electrolyte stability and the interfacial contact is of great importance in the material design of ceramic solid-state electrolytes for metal-air batteries. However, the cycling stability of 30 cycles is still far from satisfactory, and further investigations of solid-state metal-air batteries with performances acceptable for practical use are required.

Material Design of Separators

A separator is a membrane placed in the electrolyte between the electrodes to avoid direct contact of the electrodes while permitting the ionic flow. An ideal separator should be electronic insulating, inert to chemical reactions, and with no resistance

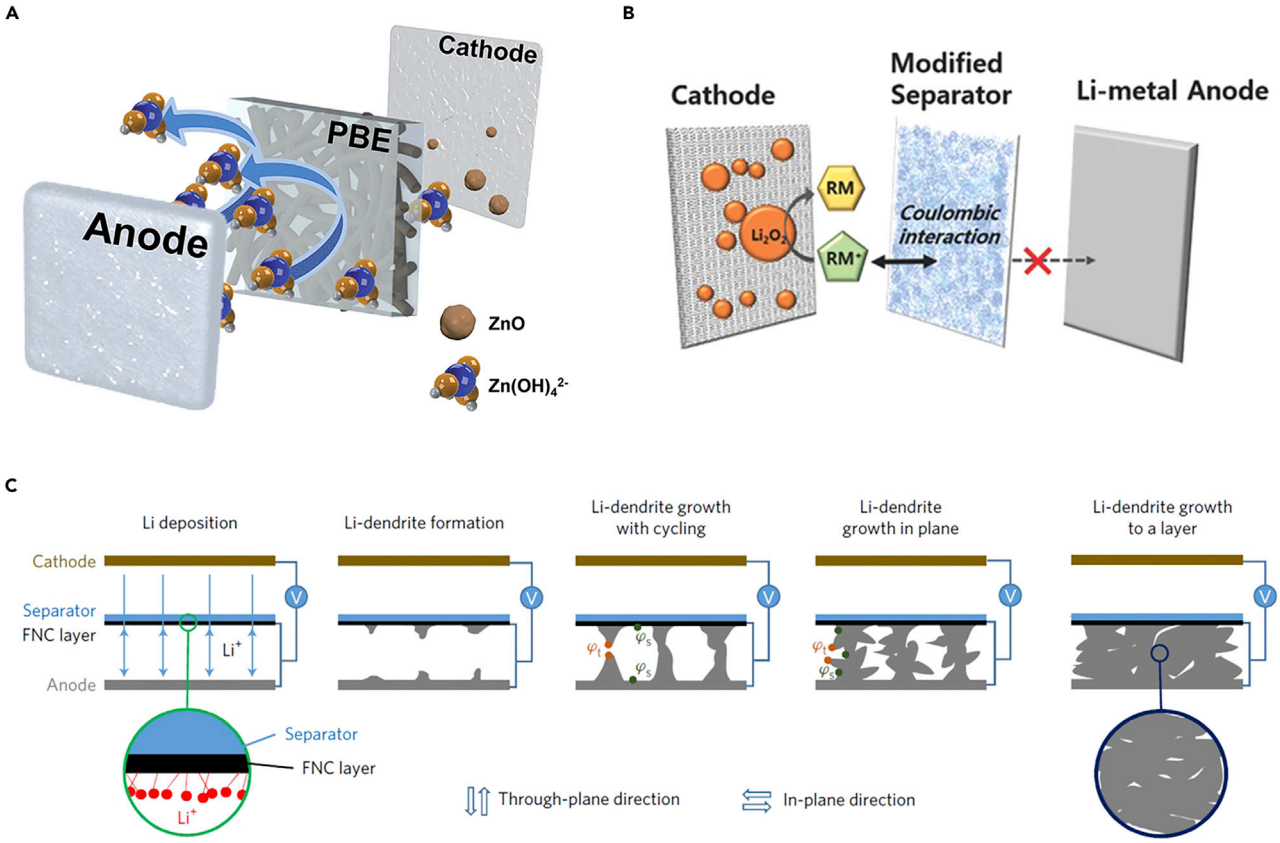


Figure 12. Material Design of Separators to Improve the Performance of Metal-Air Batteries

(A) Conceptual illustration of the role of PBE separator on mitigating zincate crossover in Zn-air batteries. Reproduced with permission from Kim et al.[158] Copyright 2016, Royal Society of Chemistry.

(B) Schematic illustration of a modified separator to suppress the migration of redox mediators in Li-O_2 batteries. Reproduced with permission from Lee et al.[160] Copyright 2017, Wiley-VCH Verlag GmbH & Co. KGaA.

(C) Li dendrite growth in a cell with FNC-coated separator. φ_s and φ_t represent the potentials at the base and the tip of the dendrite, respectively. The potential difference of φ_t and φ_s is the driving force for the dendrite growth. Reproduced with permission from Liu et al.[161] Copyright 2017, Springer Nature.

for ionic transport, while separators in reality could only be close to the ideal status, with acceptable electronic/ionic resistivity and chemical stability.[154] Materials for battery separators include woven fibers, polymer films, and solid ion conductors.[154] Besides the basic functions, a separator can also improve the metal-air battery performance by limiting the migration of some specific ions or molecules.[155–161]

In Zn-air batteries, the migration of zincate ions to the air cathode is responsible for the loss of Coulombic efficiency and also influences the activity of oxygen electrocatalysts. Therefore, separators that suppress zincate crossover have been proposed to solve this problem.[157,158] Kim et al.[158] reported a polymer blend electrolyte (PBE) membrane composed of ion-conducting PVA/PAA (PAA = polyacrylic acid) and ion-repelling Nafion as the separator of Zn-air batteries. The Nafion phase can limit the diffusion of zincate ions by the Donnan exclusion effect, thus minimizing the formation of ZnO on the air electrode, as illustrated in Figure 12A. The Zn-air battery with the PBE separator showed electrochemical rechargeability over 2,500 min at 20 mA cm^{-2}. In contrast, the Zn-air battery with commercial Celgard3501 separator reached the cutoff voltage after only 900 min. For Li-air batteries, to prevent the decomposition of redox mediators on the Li electrode, one solution is to apply

an RM-sieving separator to restrain the shuttling.[159,160] Lee et al.[160] modified a commercial glass fiber (GF/C) separator with a polymer mixture of PEDOT:PSS (poly(3,4-ethylenedioxythiophene)polystyrene sulfonate) to suppress the migration of the redox mediator DMPZ (5,10-dihydro-5,10-dimethylphenazine) in Li-O_2 batteries. The role of this modified separator is illustrated in Figure 12B. The negatively charged groups in the separator interact with $DMPZ^+$ by Coulombic interactions, thus preventing $DMPZ^+$ from diffusing to the Li electrode. As a result, the Li-O_2 battery with this modified separator and high concentration of DMPZ exhibited stable cycling performance of 90% round-trip efficiency for more than 20 cycles. The modification on the separator was also reported to control the dendrite growth of metal electrodes. Liu et al.[161] designed a unique method of coating functionalized nanocarbon (FNC) on the commercial Celgard separator of Li metal batteries to prevent the short-circuit caused by dendrites. The schematic illustration of the battery configuration is shown in Figure 12C. The FNC layer is immobilized with Li^+ ions and electrically connected to Li electrode. When charging, Li dendrites grow from both the Li electrode and the FNC layer, in the through-plane direction. With cycling, the dendrites from the opposite directions contact with each other and then grow in the in-plane-direction instead of piercing through the separator. Finally, a dense Li metal layer is formed between the Li anode and the separator. In Li/Li symmetric cells with various types of electrodes, the cells with FNC-coated separators exhibited improved cycle life and Coulombic efficiency compared with the blank cells with clean Celgard separators. This research provides a promising approach to suppress the Li dendrite growth in Li-air batteries, and also broadens the possibilities of the material design of separators.

Summary and Outlook

Metal-air batteries hold a significant position in the research of next-generation energy-storage devices because of their ultrahigh energy density. However, the practical application of rechargeable metal-air batteries still faces challenges such as the ORR/OER overpotential, metal electrode reversibility, and electrode and electrolyte stability. In this review we have summarized the material design targets and strategies of the air electrode, metal electrode, electrolyte, and separator of metal-air batteries. The material designs were conducted from the following aspects: the oxygen electrocatalyst, pore structure, and GDL of the air electrode; the electrode composition, additive, and protective layer of the metal electrode; the phase state, composition, and additive of the electrolyte; and the selective permeability of the separator. Through these modifications of materials for metal-air batteries, the discharge power density, capacity, round-trip efficiency, Coulombic efficiency, and cycle life can be improved.

The development of rechargeable metal-air batteries has been attracting increasing attention in recent years. More insights into the scientific and technical issues in metal-air batteries are required to boost the practical application of metal-air batteries. For the future research on metal-air batteries, here we present some personal perspectives for reference.

(1) *More integrated studies*. Research on oxygen electrocatalysts makes up the majority of the research on metal-air batteries. Sometimes the metal-air battery is just an evaluation method for the oxygen electrocatalysts. However, for the sake of the development of metal-air battery technology, the whole cell system should be considered together, and attention must be paid to achieving some performance indicators for practical applications.

(2) *Rechargeability*. The low rechargeability of existing metal-air batteries mainly manifest in the round-trip efficiency and the Coulombic efficiency. A low round-trip efficiency makes a metal-air battery highly energy-consuming and uneconomical. A low Coulombic efficiency leads to quick capacity loss in a metal-air battery. These significantly hinder the practical use of rechargeable metal-air batteries. Research on the material design of metal-air batteries should preferentially address the issue of rechargeability by improving metal electrode reversibility and suppressing side reactions.

(3) *Fundamental understanding*. Effective material design strategies of metal-air batteries rely on the fundamental understanding of the electrode reactions, side reactions, dendrite growth, and so forth. Therefore, progress in advanced experimental technologies and theoretical studies can benefit the design of proper materials for metal-air batteries.

(4) *Finding suitable applications*. Metal-air batteries have significant advantages regarding energy density, but also carry certain drawbacks regarding rechargeability. Besides developing material design methods to improve the battery performance, finding suitable applications, highly desirable for high-capacity energy storage, is also crucial for realizing the practical deployment of metal-air batteries.

(5) *Technical designs*. This review focuses on the material design of metal-air batteries. Meanwhile there are also some investigations focusing on the technical designs of the cell configuration, for example, the construction of flow batteries with circulating electrolyte. A proper cell configuration is expected to take full advantage of the rationally designed materials for metal-air batteries.

Developing efficient metal-air batteries needs the rational design of materials of the air electrode, metal electrode, electrolyte, and separator. Comprehensive understanding of materials of each part of metal-air batteries requires the integration of interdisciplinary knowledge of electrochemistry, inorganic chemistry, organic chemistry, catalysis, material science, and engineering. Therefore, collaborations between researchers with different backgrounds are anticipated to overcome critical scientific and technical problems of metal-air batteries and further promote their practical applications.

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AUTHOR CONTRIBUTIONS

Conceptualization, Q.X.; Writing – Original Draft, H.-F.W.; Writing – Review & Editing, Q.X. and H.-F.W.; Supervision, Q.X.

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